

Sid Mathur's Thematic ETF Screener – Daily Report

2026-08-22

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Action Center

Run date: 2026-08-22 **Universe:** 659 ETFs / 26 clusters

Full PC1: LONG since 2026-08-04 (D+13, 0.3%) **Clusters long:** 13/26 **ETFs long:** 361/659 (55%)

Today's events (cluster level)

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

SELL (1)

cluster	theme	spread_sell	p_h5	nss
cluster_16	Municipal Bond	-0.183	0.51	-0.514

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_11	Healthcare	7.248	1	1.00	1.000	1.00
cluster_9	U.S. Dividend Equity	4.104	1	1.00	1.000	0.96
cluster_3	U.S. Large Cap Value	3.106	1	1.00	0.978	0.88
cluster_2	U.S. Mid Cap Equity	1.374	1	0.94	0.609	0.56
cluster_14	Financials	3.187	1	1.00	0.201	0.24

Imminent SELL (top 1 by h=1 NSS)

cluster	theme	spread_sell	p_h1	p_h5	nss_h1	nss_h5
cluster_18	Consumer Staples	-0.332	0.877	0.74	-0.749	-0.58

Full Sample PC1

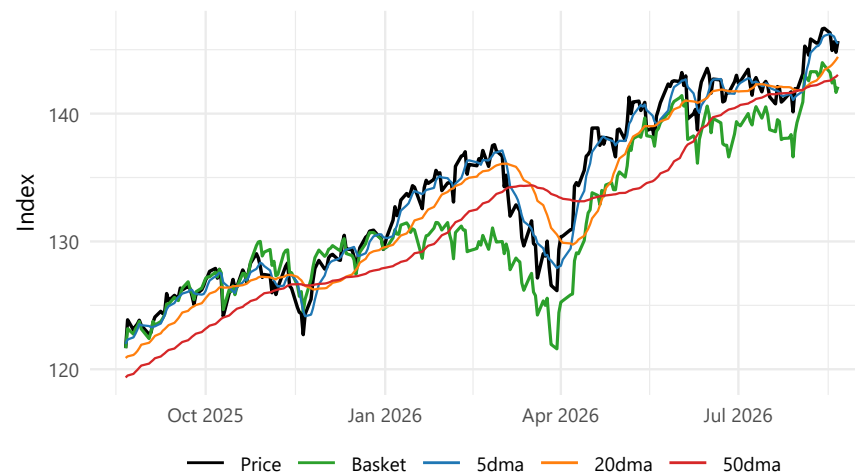


Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: LONG since 2026-08-04 (13 days, 0.3%)

5dma: 145.469 20dma: 144.439 50dma: 143.031

Spread vs 50dma: +2.438 Spread vs 20dma: +1.030

P(buy crossover within 5 days): 100.0% P(sell): 3.5%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
SRLN	cluster_23	Bank Loan / Floating Rate	1.000	4	0.1%	5.98%	0.89
FLOT	cluster_23	Bank Loan / Floating Rate	1.000	4	0.1%	5.86%	0.89
GUNR	cluster_12	Materials	1.000	12	5.1%	5.39%	0.98
BKLN	cluster_23	Bank Loan / Floating Rate	1.000	4	0.1%	4.34%	0.89
FLRN	cluster_23	Bank Loan / Floating Rate	1.000	4	0.1%	2.46%	0.89
IAU	cluster_17	Gold	1.000	10	8.9%	2.22%	0.95
SLV	cluster_17	Gold	1.000	10	8.9%	1.21%	0.95
AMLX	cluster_12	Materials	1.000	12	5.1%	0.97%	0.98
FLTR	cluster_23	Bank Loan / Floating Rate	1.000	4	0.1%	0.70%	0.89
URA	cluster_12	Materials	1.000	12	5.1%	0.67%	0.98
GLD	cluster_17	Gold	1.000	10	8.9%	0.08%	0.95
VEA	cluster_4	Developed ex-US Equity	0.964	16	2.3%	8.12%	0.99
PDBC	cluster_19	Broad Commodities	0.957	8	4.2%	5.17%	0.98
IEMG	cluster_8	Emerging Markets Equity	0.952	6	0.3%	4.32%	0.99
ICSH	cluster_24	Cash / Ultra-Short Fixed In...	0.925	4	0.1%	7.97%	0.86
SHV	cluster_24	Cash / Ultra-Short Fixed In...	0.925	4	0.1%	5.09%	0.86
MINT	cluster_24	Cash / Ultra-Short Fixed In...	0.925	4	0.1%	3.46%	0.86
GBIL	cluster_24	Cash / Ultra-Short Fixed In...	0.925	4	0.1%	2.51%	0.86
VOO	cluster_1	U.S. Large Cap Equity	0.850	12	-0.7%	7.91%	0.99
BBAX	cluster_7	Asia-Pacific Equity	0.793	27	2.9%	5.65%	0.97
IGF	cluster_7	Asia-Pacific Equity	0.793	27	2.9%	3.20%	0.97
EWZ	cluster_7	Asia-Pacific Equity	0.793	27	2.9%	0.99%	0.97
QQQ	cluster_6	U.S. Large Cap Growth	0.780	10	-1.5%	4.57%	0.98
EWJ	cluster_22	Japan Equity	0.763	11	0.3%	5.14%	0.97
KWEB	cluster_13	China Equity	0.331	14	0.2%	3.77%	0.92
XLP	cluster_18	Consumer Staples	-0.581	3	-0.6%	5.33%	0.99
XLU	cluster_18	Consumer Staples	-0.581	3	-0.6%	3.29%	0.99

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1.00	0.243	1.095	0.96
VTV	cluster_3	U.S. Large Cap Value	1.00	0.878	1.092	0.97
VHT	cluster_11	Healthcare	1.00	1.000	1.088	0.93
SCHD	cluster_9	U.S. Dividend Equity	1.00	0.956	0.891	0.96
BWX	cluster_25	International Fixed Income	1.00	0.990	0.752	0.93
INDA	cluster_20	India Equity	0.99	0.572	0.660	0.91
NOBL	cluster_20	India Equity	0.99	0.572	0.080	0.91
PPLT	cluster_20	India Equity	0.99	0.572	0.030	0.91
IJH	cluster_2	U.S. Mid Cap Equity	0.94	0.562	1.016	0.99

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
11	cluster_11	Healthcare	Healthcare	17	FLAT				1.000	+0.000
12	cluster_12	Materials	Materials	17	LONG		12	5.1%	1.000	+0.000
17	cluster_17	Gold	Gold	12	LONG		10	8.9%	1.000	+0.000
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	LONG		4	0.1%	1.000	+0.004
25	cluster_25	International Fixed Income	International Fix...	6	FLAT				0.990	+0.014
4	cluster_4	Developed ex-US Equity	Developed ex-US E...	60	LONG		16	2.3%	0.964	-0.029
19	cluster_19	Broad Commodities	Broad Commodities	11	LONG		8	4.2%	0.957	+0.059
9	cluster_9	U.S. Dividend Equity	U.S. Dividend Equ...	25	FLAT				0.956	-0.022
8	cluster_8	Emerging Markets Equity	Emerging Markets ...	25	LONG		6	0.3%	0.952	-0.002
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	LONG		4	0.1%	0.925	+0.061
3	cluster_3	U.S. Large Cap Value	U.S. Large Cap Va...	60	FLAT				0.878	-0.102
1	cluster_1	U.S. Large Cap Equity	U.S. Large Cap Eq...	81	LONG		12	-0.7%	0.850	-0.123
7	cluster_7	Asia-Pacific Equity	Asia-Pacific Equi...	26	LONG		27	2.9%	0.793	+0.160
6	cluster_6	U.S. Large Cap Growth	U.S. Large Cap Gr...	52	LONG		10	-1.5%	0.780	-0.147
22	cluster_22	Japan Equity	Japan Equity	9	LONG		11	0.3%	0.763	-0.201
20	cluster_20	India Equity	India Equity	10	FLAT				0.572	+0.047
2	cluster_2	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	72	FLAT				0.562	-0.359
13	cluster_13	China Equity	China Equity	15	LONG		14	0.2%	0.331	-0.043
14	cluster_14	Financials	Financials	14	FLAT				0.243	-0.371
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.182	+0.147
5	cluster_5	Aggregate Fixed Income	Aggregate Fixed I...	58	FLAT				-0.305	+0.014
10	cluster_10	High Yield Bond	High Yield Bond	21	FLAT				-0.335	-0.207
16	cluster_16	Municipal Bond	Municipal Bond	13	FLAT				-0.514	-0.170
18	cluster_18	Consumer Staples	Consumer Staples	11	LONG		3	-0.6%	-0.581	-0.308
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT				-0.885	+0.075
15	cluster_15	Real Estate	Real Estate	14	FLAT				-0.945	-0.009

Cluster Details

Cluster 1 – U.S. Large Cap Equity

Members: 81 **Var explained:** 0.858 **Signal:** LONG since 2026-08-05 (12 days, -0.7%)

Signal logic:

- Latest buy_cross: 2026-08-05 – filter **PASS** (252d at date = +20.1%)
- Latest sell_cross: 2026-07-17
- Current 252d return: +18.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-05 > last sell at 2026-07-17)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1190
U.S. Total Market Equity	7	0.0930
Global Equity	7	0.0910
Technology	5	0.0601
Consumer Discretionary	5	0.0587
Asset Allocation	4	0.0483
Industrials	4	0.0408
U.S. Equity	3	0.0388

U.S. Mega Cap Equity	3	0.0387
U.S. Multifactor Equity	3	0.0387
U.S. Quality / Factor Equity	3	0.0382
U.S. Mid Cap Growth	3	0.0381
U.S. Momentum Equity	3	0.0344
Communication Services	3	0.0336
U.S. ESG Equity	2	0.0264
Momentum Equity	2	0.0238
Convertible Bond	2	0.0227
Commodity	1	0.0132
Large Cap Value Equities	1	0.0131
Large Cap Blend Equities	1	0.0131
Dividend Equity	1	0.0128
U.S. Equal Weight Equity	1	0.0126
U.S. Dividend Equity	1	0.0125
Growth Equity	1	0.0123
IPO Equity	1	0.0118
U.S. Mid Cap Momentum Equity	1	0.0117
Alternatives	1	0.0115
U.S. Small Cap Momentum Equity	1	0.0112
International IPO Equity	1	0.0101
Israel Equity	1	0.0097

Replication basket – VOO (cor=0.99) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
VOO	1.007	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
50	IWV	0.0133	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
51	IYY	0.0133	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
57	IWB	0.0133	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
62	VTHR	0.0133	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
65	VONE	0.0133	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
69	ITOT	0.0133	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
71	SCHB	0.0133	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
72	SCHX	0.0133	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
75	VTI	0.0133	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
5	SPY	0.0132	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG

12	URTH	0.0132	iShares MSCI World ETF	Global Equity	LONG
37	TILT	0.0132	FlexShares Morningstar US Market Factors...	U.S. Multifactor Equity	FLAT
46	JHML	0.0132	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
49	SUSA	0.0132	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
54	SPYX	0.0132	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
56	ESGU	0.0132	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
63	GSLC	0.0132	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
67	VV	0.0132	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
68	ILCB	0.0132	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
70	IVV	0.0132	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
73	SPTM	0.0132	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
74	VOO	0.0132	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
7	ACWI	0.0131	iShares MSCI ACWI ETF	Global Equity	LONG
13	CRBN	0.0131	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
14	SPGM	0.0131	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
15	VT	0.0131	Vanguard Total World Stock Index ETF	Global Equity	LONG
48	DSI	0.0131	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
52	LRGF	0.0131	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
58	IWL	0.0131	iShares Russell Top 200 ETF	U.S. Equity	LONG
64	EPS	0.0131	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
66	MGC	0.0131	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
8	AOA	0.0130	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
53	OEF	0.0129	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
59	QUAL	0.0129	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	FLAT
36	IMCG	0.0128	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
44	QDF	0.0128	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
45	CATH	0.0128	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
35	VOT	0.0127	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
43	IOO	0.0127	iShares Global 100 ETF	Global Equity	LONG
55	XLG	0.0127	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
60	QUS	0.0127	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
11	AOR	0.0126	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
18	QQEW	0.0126	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
19	XT	0.0126	iShares Future Exponential Technologies...	Technology	LONG
28	IWP	0.0126	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
61	SPHQ	0.0126	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
21	FTC	0.0125	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
42	SPHB	0.0125	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
47	DGRW	0.0125	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
81	OMFL	0.0124	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
26	RPG	0.0123	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
4	ROBO	0.0122	ROBO Global Robotics and Automation Ind...	Technology	LONG
24	IYC	0.0122	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
22	PDP	0.0121	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
31	VCR	0.0120	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG

34	FDIS	0.0120	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
77	VFMO	0.0120	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
3	BOTZ	0.0119	Global X Robotics & Artificial Intellig...	Technology	LONG
1	XLY	0.0118	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
10	AOM	0.0118	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
23	FPX	0.0118	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
16	FV	0.0117	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
27	XMMO	0.0117	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
25	CWB	0.0116	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
6	QAI	0.0115	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
30	MTUM	0.0115	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
32	VOX	0.0113	Vanguard Communication Services ETF	Communication Services	LONG
33	FCOM	0.0113	Fidelity MSCI Communication Services In...	Communication Services	LONG
38	DWAS	0.0112	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	LONG
29	ICVT	0.0111	iShares Convertible Bond ETF	Convertible Bond	LONG
17	PRNT	0.0110	3D Printing (The) ETF	Technology	LONG
2	XLC	0.0109	State Street Communication Services Sel...	Communication Services	LONG
9	AOK	0.0109	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
80	PRN	0.0109	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
76	QMOM	0.0108	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
79	RTH	0.0107	VanEck Retail ETF	Consumer Discretionary	FLAT
39	PPA	0.0102	Invesco Aerospace & Defense ETF	Industrials	LONG
41	XAR	0.0102	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG
20	FPXI	0.0101	First Trust International Equity Opport...	International IPO Equity	FLAT
78	EIS	0.0097	iShares MSCI Israel ETF	Israel Equity	LONG
40	ITA	0.0095	iShares U.S. Aerospace & Defense ETF	Industrials	LONG

Cluster 2 – U.S. Mid Cap Equity

Members: 72 **Var explained:** 0.897 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +40.0%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +23.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
U.S. Small Cap Equity	10	0.1440
U.S. Mid Cap Equity	6	0.0863
U.S. Small Cap Value	6	0.0859
U.S. Small Cap Growth	6	0.0841
Consumer Discretionary	4	0.0517
Industrials	4	0.0498
U.S. Mid Cap Value	3	0.0432
U.S. Mid Cap Growth	3	0.0420
Leveraged / Inverse	2	0.0276
Dividend Equity	2	0.0273
U.S. Small Cap Low Volatility	2	0.0261
U.S. Small Cap Fundamental Equity	1	0.0146
U.S. Small Cap ESG Equity	1	0.0145
Unknown	1	0.0145
U.S. Mid/Small Cap Equity	1	0.0145
U.S. Small/Mid Cap Fundamental Equity	1	0.0145
Mid Cap Value Equities	1	0.0144
U.S. Small Cap Multifactor Equity	1	0.0144
Small Cap Value Equities	1	0.0142
U.S. Small/Mid Cap Equity	1	0.0142
U.S. Mid Cap Dividend Equity	1	0.0141
U.S. ESG Equity	1	0.0141

U.S. Extended Market Equity	1	0.0141
U.S. Small Cap Dividend Equity	1	0.0139
Value Equity	1	0.0139
U.S. Equity	1	0.0139
U.S. Mid Cap Quality Equity	1	0.0139
Growth Equity	1	0.0139
U.S. Small Cap Industrials	1	0.0136
U.S. Small Cap Momentum Equity	1	0.0136
U.S. Micro Cap Equity	1	0.0134
U.S. Small Cap Consumer Discretionary	1	0.0131
U.S. Dividend Equity	1	0.0131
U.S. Mid Cap Low Volatility	1	0.0124
Real Estate	1	0.0113

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.013

ticker	rep_wt	name	category	etf_signal
IJH	1.016	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
47	FNDA	0.0146	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
8	FNX	0.0145	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
11	NUSC	0.0145	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
12	PRFZ	0.0145	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
15	MDY	0.0145	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
23	IVOO	0.0145	Vanguard S&P Mid-Cap 400 ETF		FLAT
24	VI00	0.0145	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
27	IJR	0.0145	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
30	IJH	0.0145	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
31	SPMD	0.0145	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
32	SPSM	0.0145	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
33	VB	0.0145	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
59	VBR	0.0145	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
4	IWM	0.0144	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
9	FYX	0.0144	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
13	SMLF	0.0144	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
16	IJT	0.0144	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
18	VIOG	0.0144	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
21	SLYG	0.0144	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT

25	VTWO	0.0144	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
34	SCHA	0.0144	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
44	EZM	0.0144	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
50	IWN	0.0144	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
52	IJJ	0.0144	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
54	IVOV	0.0144	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
55	MDYV	0.0144	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
5	JHMM	0.0143	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
7	SCHM	0.0143	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
53	IJS	0.0143	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
56	SLYV	0.0143	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
58	VTWV	0.0143	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
65	RWK	0.0143	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
43	EES	0.0142	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
45	PSC	0.0142	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
57	VIOV	0.0142	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
61	RWJ	0.0142	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
71	SDVY	0.0142	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	FLAT
72	ISMD	0.0142	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
29	VXF	0.0141	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
42	DON	0.0141	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Equ...	FLAT
60	ETHO	0.0141	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
17	IJK	0.0140	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
19	IVOG	0.0140	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
20	MDYG	0.0140	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
28	ISCG	0.0139	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
41	DES	0.0139	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
63	XMHQ	0.0139	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
64	RFV	0.0139	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
66	OMFS	0.0139	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
70	FYT	0.0139	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
14	IWO	0.0137	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
22	VTWG	0.0137	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
67	RZV	0.0137	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
26	VBK	0.0136	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
62	PSCI	0.0136	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
69	XSMO	0.0136	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
35	FXD	0.0135	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	FLAT
10	IWC	0.0134	iShares Microcap ETF	U.S. Micro Cap Equity	LONG
39	RSPD	0.0132	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
51	SMMV	0.0132	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
6	REGL	0.0131	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	FLAT
37	FTXR	0.0131	First Trust Nasdaq Transportation ETF	Industrials	FLAT
40	SMDV	0.0131	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
68	PSCD	0.0131	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT

46	XTN	0.0129	State Street SPDR S&P Transportation ETF	Industrials	FLAT
49	XSLV	0.0128	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
1	XRT	0.0126	State Street SPDR S&P Retail ETF	Consumer Discretionary	FLAT
36	PEJ	0.0124	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
38	IYT	0.0124	iShares U.S. Transportation ETF	Industrials	LONG
48	XMLV	0.0124	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
2	JETS	0.0114	U.S. Global Jets ETF	Industrials	FLAT
3	REM	0.0113	iShares Mortgage Real Estate ETF	Real Estate	LONG

Cluster 3 – U.S. Large Cap Value

Members: 60 Var explained: 0.890 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +26.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +20.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Value	10	0.1728
U.S. Dividend Equity	8	0.1359
Industrials	7	0.1142
Materials	4	0.0627
U.S. Mid Cap Equity	3	0.0516
U.S. Multifactor Equity	3	0.0512
U.S. Value Factor Equity	3	0.0502
Utilities	3	0.0468
U.S. Fundamental Equity	2	0.0348
U.S. Equal Weight Equity	2	0.0347
U.S. Mid Cap Value	2	0.0346
U.S. Equity	2	0.0343
Consumer Discretionary	2	0.0272
Large Cap Blend Equities	1	0.0173
U.S. Size Factor Equity	1	0.0172
U.S. Revenue Weighted Equity	1	0.0171
Value Equity	1	0.0170
U.S. Mega Cap Value	1	0.0169
U.S. Buyback Equity	1	0.0168
U.S. Large Cap Equity	1	0.0167
U.S. Moat Equity	1	0.0162
Communication Services	1	0.0137

Replication basket – VTV (cor=0.97) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
VTV	1.092	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
3	RSP	0.0175	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
26	IWD	0.0175	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
36	VONV	0.0175	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
15	PRF	0.0174	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
18	CFA	0.0174	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
22	FNDX	0.0174	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
29	JPUS	0.0174	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT

42	SCHV	0.0174	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
48	IWS	0.0174	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
6	FEX	0.0173	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
39	ILCV	0.0173	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
41	IUSV	0.0173	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
59	JPME	0.0173	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
23	EQAL	0.0172	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
27	IWR	0.0172	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
28	IVE	0.0172	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
30	SIZE	0.0172	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	FLAT
34	VOOV	0.0172	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	FLAT
40	IMCB	0.0172	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
43	SPYV	0.0172	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
45	VTV	0.0172	Vanguard Value ETF	U.S. Large Cap Value	LONG
49	VOE	0.0172	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
52	DTD	0.0172	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
16	RWL	0.0171	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
20	NULV	0.0171	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
44	VO	0.0171	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
54	DGRO	0.0171	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
60	JVAL	0.0171	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
12	IYJ	0.0170	iShares U.S. Industrials ETF	Industrials	FLAT
17	AIVL	0.0170	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
24	IWX	0.0170	iShares Russell Top 200 Value ETF	Value Equity	LONG
25	ONEV	0.0170	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
38	VYM	0.0170	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
50	PFM	0.0170	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
51	DLN	0.0170	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
55	VIG	0.0170	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
37	MGV	0.0169	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	FLAT
5	PKW	0.0168	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
10	RDVY	0.0168	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
21	FDVV	0.0168	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
4	FXR	0.0167	First Trust Industrials AlphaDEX	Industrials	FLAT
53	DIA	0.0167	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	FLAT
33	VIS	0.0166	Vanguard Industrials ETF	Industrials	LONG
35	FIDU	0.0166	Fidelity MSCI Industrials Index ETF	Industrials	LONG
1	XLI	0.0164	State Street Industrial Select Sector S...	Industrials	FLAT
19	CFO	0.0164	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
11	MOAT	0.0162	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
56	PAVE	0.0162	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
31	VLUE	0.0161	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
9	FIW	0.0160	First Trust Water ETF	Utilities	FLAT
32	VAW	0.0160	Vanguard Materials ETF	Materials	LONG
2	XLB	0.0158	State Street Materials Select Sector SP...	Materials	LONG

7	PHO	0.0158	Invesco Water Resources ETF	Utilities	FLAT
14	RSPM	0.0156	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
13	IYM	0.0152	iShares U.S. Basic Materials ETF	Materials	LONG
8	CGW	0.0150	Invesco S&P Global Water Index ETF	Utilities	FLAT
58	PKB	0.0148	Invesco Building & Construction ETF	Industrials	FLAT
47	XHB	0.0142	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
57	IYZ	0.0137	iShares U.S. Telecommunications ETF	Communication Services	LONG
46	ITB	0.0130	iShares U.S. Home Construction ETF	Consumer Discretionary	LONG

Cluster 4 – Developed ex-US Equity

Members: 60 **Var explained:** 0.897 **Signal:** LONG since 2026-07-30 (16 days, 2.3%)

Signal logic:

- Latest buy_cross: 2026-07-30 – filter **PASS** (252d at date = +22.0%)
- Latest sell_cross: 2026-06-26
- Current 252d return: +20.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-30 > last sell at 2026-06-26)



Category breakdown

category	n_members	abs_weight
International Equity	19	0.3204
Developed ex-US Equity	9	0.1551
Global ex-US Equity	5	0.0861
Europe Equity	4	0.0685
International Dividend Equity	4	0.0668
Alternatives	4	0.0625
Eurozone Equity	3	0.0497
Commodity	1	0.0174
Global ex-US Small Cap Equity	1	0.0168
Germany Equity	1	0.0164
France Equity	1	0.0164
Italy Equity	1	0.0161
Europe Financials	1	0.0160
United Kingdom Equity	1	0.0160
Netherlands Equity	1	0.0159
Sweden Equity	1	0.0158
Spain Equity	1	0.0153
Austria Equity	1	0.0150
Poland Equity	1	0.0137

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.022

ticker	rep_wt	name	category	etf_signal
VEA	0.958	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0175	iShares MSCI EAFE ETF	Developed ex-US Equity	LONG
3	VEA	0.0175	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
41	ESGD	0.0175	iShares ESG Aware MSCI EAFE ETF	International Equity	LONG
51	IEFA	0.0175	iShares Core MSCI EAFE ETF	Developed ex-US Equity	LONG
40	GSIE	0.0174	Goldman Sachs ActiveBeta International ...	Commodity	LONG
52	SCHF	0.0174	Schwab International Equity ETF	Developed ex-US Equity	LONG
53	SPDW	0.0174	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
16	DWM	0.0173	WisdomTree International Equity Fund	International Dividend E...	LONG
25	TLTD	0.0173	FlexShares Morningstar Developed Market...	International Equity	LONG
32	INTF	0.0173	iShares International Equity Factor ETF	International Equity	LONG

46	IEUR	0.0173	iShares Core MSCI Europe ETF	Europe Equity	LONG
47	IXUS	0.0173	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
49	VGK	0.0173	Vanguard FTSEEuropean ETF	Europe Equity	LONG
6	IEV	0.0172	iShares Europe ETF	Europe Equity	LONG
19	PXF	0.0172	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
29	ACWX	0.0172	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
33	IQLT	0.0172	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
39	FNDF	0.0172	Schwab Fundamental International Equity...	International Equity	LONG
48	VEU	0.0172	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
50	VXUS	0.0172	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
30	CWI	0.0171	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
35	QEFA	0.0171	State Street SPDR MSCI EAFE StrategicFa...	International Equity	LONG
14	EZU	0.0170	iShares MSCI Eurozone ETF	Eurozone Equity	LONG
22	SCZ	0.0170	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
23	EFV	0.0170	iShares MSCI EAFE Value ETF	Developed ex-US Equity	LONG
26	JPIN	0.0170	JPMorgan Diversified Return Internation...	International Equity	LONG
38	VYMI	0.0170	Vanguard International High Dividend Yi...	International Dividend E...	LONG
44	SCHC	0.0170	Schwab International Small-Cap Equity E...	Developed ex-US Equity	LONG
18	IQDF	0.0169	FlexShares International Quality Divide...	International Equity	LONG
24	FNDC	0.0169	Schwab Fundamental International Small ...	International Equity	LONG
34	IVLU	0.0169	iShares MSCI Intl Value Factor ETF	International Equity	LONG
36	FEZ	0.0169	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	LONG
37	RODM	0.0169	Hartford Multifactor Developed Markets ...	International Equity	LONG
56	HFXI	0.0169	NYLI FTSE International Equity Currency...	International Equity	LONG
7	DLS	0.0168	WisdomTree International SmallCap Fund	International Equity	LONG
20	EFG	0.0168	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	LONG
45	VSS	0.0168	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
4	FEP	0.0167	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
31	IMTM	0.0166	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
43	VIGI	0.0166	Vanguard International Dividend Appreci...	International Dividend E...	LONG
1	EWG	0.0164	iShares MSCI Germany Index Fund	Germany Equity	LONG
12	EWQ	0.0164	iShares MSCI France Index Fund	France Equity	LONG
21	GWX	0.0164	State Street SPDR S&P International Sma...	International Equity	LONG
5	VIDI	0.0162	Vident International Equity Strategy ETF	International Equity	LONG
59	IPKW	0.0162	Invesco International BuyBack Achievers...	International Equity	LONG
60	EWI	0.0161	iShares MSCI Italy ETF	Italy Equity	FLAT
13	EWU	0.0160	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
15	IDV	0.0160	iShares International Select Dividend E...	International Dividend E...	LONG
17	EUFN	0.0160	iShares MSCI Europe Financials ETF	Europe Financials	LONG
27	DBEF	0.0160	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
10	HEZU	0.0159	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	LONG
28	HEFA	0.0159	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
58	EWN	0.0159	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
42	HDEF	0.0158	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
55	EWD	0.0158	iShares MSCI Sweden ETF	Sweden Equity	LONG

8	HEDJ	0.0154	WisdomTree Europe Hedged Equity Fund	Alternatives	LONG
9	IHDG	0.0153	WisdomTree International Hedged Quality...	Alternatives	FLAT
11	EWP	0.0153	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWO	0.0150	iShares MSCI Austria ETF	Austria Equity	LONG
54	EPOL	0.0137	iShares MSCI Poland ETF	Poland Equity	LONG

Cluster 5 – Aggregate Fixed Income

Members: 58 **Var explained:** 0.831 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +0.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3284

Investment Grade Corporate Bond	13	0.2303
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0891
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0306
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.006

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	FLAT
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
46	SCHR	0.0183	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	FLAT
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
27	USIG	0.0181	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
50	MBB	0.0181	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	FLAT
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT
52	SPMB	0.0180	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT

3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	FLAT
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
10	CORP	0.0178	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
16	SPLB	0.0172	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	FLAT
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	FLAT
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	FLAT
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	LONG
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	LONG
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	LONG
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	LONG
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	FLAT
33	BSCQ	0.0155	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	LONG
14	IBDR	0.0150	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	LONG
38	CMBS	0.0135	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	LONG
32	LDUR	0.0121	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 6 – U.S. Large Cap Growth

Members: 52 **Var explained:** 0.837 **Signal:** LONG since 2026-08-07 (10 days, -1.5%)

Signal logic:

- Latest buy_cross: 2026-08-07 – filter **PASS** (252d at date = +28.5%)
- Latest sell_cross: 2026-07-17
- Current 252d return: +27.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-07 > last sell at 2026-07-17)



Category breakdown

category	n_members	abs_weight
Technology	29	0.5520
U.S. Large Cap Growth	11	0.2258
Growth Equity	2	0.0403
Financials	2	0.0347
U.S. Equity	1	0.0207
U.S. Mega Cap Growth	1	0.0204
Options Income	1	0.0185
IPO Equity	1	0.0183
U.S. Small Cap Technology	1	0.0183
Trend-Following Equity	1	0.0181
Consumer Discretionary	1	0.0171
U.S. Trend-Following Equity	1	0.0158

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.022

ticker	rep_wt	name	category	etf_signal
QQQ	1.059	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
30	IGM	0.0208	iShares Expanded Tech Sector ETF	Technology	LONG
12	QQQ	0.0207	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
34	ONEQ	0.0207	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
43	ILCG	0.0207	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
32	IYW	0.0206	iShares U.S. Technology ETF	Technology	LONG
36	IWF	0.0206	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
38	VGT	0.0206	Vanguard Information Tech ETF	Technology	LONG
40	FTEC	0.0206	Fidelity MSCI Information Technology In...	Technology	LONG
41	VONG	0.0206	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
47	VUG	0.0206	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
24	XNTK	0.0205	State Street SPDR NYSE Technology ETF	Technology	LONG
33	NULG	0.0205	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
44	IUSG	0.0205	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
14	QTEC	0.0204	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
35	IWY	0.0204	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
37	IVW	0.0204	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
39	VOOG	0.0204	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
42	MGK	0.0204	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
45	SCHG	0.0204	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
46	SPYG	0.0204	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
1	XLK	0.0203	State Street Technology Select Sector S...	Technology	LONG
19	FXL	0.0203	First Trust Technology AlphaDEX	Technology	LONG
31	IXN	0.0203	iShares Global Tech ETF	Technology	LONG
6	AIQ	0.0202	Global X Artificial Intelligence & Tech...	Technology	LONG
29	PWB	0.0199	Invesco Large Cap Growth ETF	Growth Equity	LONG
16	TDIV	0.0193	First Trust NASDAQ Technology Dividend ...	Technology	LONG
52	QTUM	0.0191	Defiance Quantum ETF	Technology	LONG
3	SMH	0.0190	VanEck Semiconductor ETF	Technology	FLAT
23	FDN	0.0190	First Trust DJ Internet Index Fund	Technology	LONG
2	SOXX	0.0187	iShares PHLX SOX Semiconductor Sector I...	Technology	FLAT
5	SKYY	0.0186	First Trust Cloud Computing ETF	Technology	LONG
11	ARKQ	0.0186	ARK Autonomous Technology & Robotics ETF	Technology	LONG
17	XSD	0.0185	State Street SPDR S&P Semiconductor ETF	Technology	FLAT
22	PNQI	0.0185	Invesco Nasdaq Internet ETF	Technology	FLAT
28	QYLD	0.0185	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
13	NXTG	0.0184	First Trust Indxx NextG ETF	Technology	LONG

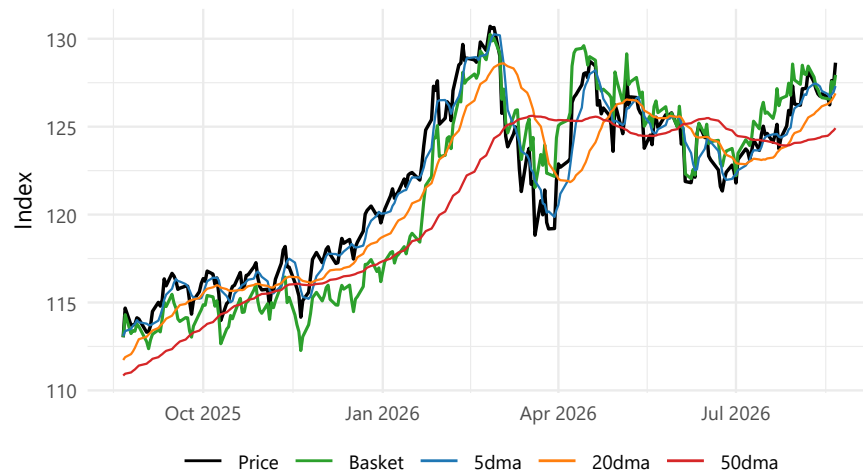
21	IPO	0.0183	Renaissance IPO ETF	IPO Equity	LONG
49	PSCT	0.0183	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	FLAT
4	IGV	0.0182	iShares Expanded Tech-Software Sector E...	Technology	FLAT
15	PSI	0.0182	Invesco Semiconductors ETF	Technology	FLAT
50	FTXL	0.0182	First Trust Nasdaq Semiconductor ETF	Technology	FLAT
51	PTF	0.0182	Invesco Dorsey Wright Technology Moment...	Technology	LONG
10	ARKW	0.0181	ARK Next Generation Internet ETF	Technology	LONG
27	PTNQ	0.0181	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
26	FINX	0.0179	Global X FinTech ETF	Financials	FLAT
9	ARKK	0.0178	ARK Innovation ETF	Technology	LONG
20	CIBR	0.0175	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
8	HACK	0.0171	Amplify Cybersecurity ETF	Technology	FLAT
18	IBUY	0.0171	Amplify Online Retail ETF	Consumer Discretionary	FLAT
25	IPAY	0.0167	Amplify Digital Payments ETF	Financials	FLAT
7	BLOK	0.0164	Amplify Blockchain Technology ETF	Technology	FLAT
48	PTLC	0.0158	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG

Cluster 7 – Asia-Pacific Equity

Members: 26 **Var explained:** 0.669 **Signal:** LONG since 2026-07-15 (27 days, 2.9%)

Signal logic:

- Latest buy_cross: 2026-07-15 – filter **PASS** (252d at date = +13.4%)
- Latest sell_cross: 2026-06-22
- Current 252d return: +13.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-15 > last sell at 2026-06-22)



Category breakdown

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0880
International Equity	2	0.0857
International Dividend Equity	2	0.0854
Real Estate	2	0.0822
Global Equity	2	0.0816
Emerging Markets Fixed Income	2	0.0747
Currency	2	0.0717
Australia Equity	1	0.0435
Canada Equity	1	0.0413
Switzerland Equity	1	0.0388
Belgium Equity	1	0.0383
Singapore Equity	1	0.0372
South Africa Equity	1	0.0367
New Zealand Equity	1	0.0351
Latin America Equity	1	0.0350
Mexico Equity	1	0.0333
Consumer Discretionary	1	0.0323
Chile Equity	1	0.0296
Brazil Equity	1	0.0295

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
EWZ	0.084	iShares MSCI Brazil ETF	Brazil Equity	FLAT
IGF	0.274	iShares Global Infrastructure ETF	Global Equity	FLAT
BBAX	0.484	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
9	EPP	0.0440	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	LONG
26	BBAX	0.0439	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
13	IDLV	0.0437	Invesco S&P International Developed Low...	International Equity	LONG
5	EWA	0.0435	iShares MSCI Australia Index Fund	Australia Equity	LONG
12	DWX	0.0431	State Street SPDR S&P International Div...	International Dividend E...	LONG
4	PID	0.0423	Invesco International Dividend Achiever...	International Dividend E...	FLAT
14	EFAV	0.0420	iShares MSCI EAFE Min Vol Factor ETF	International Equity	LONG
15	VNQI	0.0416	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
10	NFRA	0.0414	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
8	EWC	0.0413	iShares MSCI Canada Index Fund	Canada Equity	LONG
3	RWX	0.0406	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
11	IGF	0.0402	iShares Global Infrastructure ETF	Global Equity	FLAT
6	EWL	0.0388	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
25	EWK	0.0383	iShares MSCI Belgium ETF	Belgium Equity	LONG
21	FXA	0.0382	Invesco CurrencyShares Australian Dolla...	Currency	LONG
16	EBND	0.0375	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
7	EWS	0.0372	iShares MSCI Singapore ETF	Singapore Equity	LONG
17	EMLC	0.0372	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
23	EZA	0.0367	iShares MSCI South Africa Index Fund	South Africa Equity	LONG
22	ENZL	0.0351	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
20	ILF	0.0350	iShares Latin America 40 ETF	Latin America Equity	FLAT
24	FXC	0.0335	Invesco CurrencyShares Canadian Dollar ...	Currency	LONG
18	EWX	0.0333	iShares MSCI Mexico ETF	Mexico Equity	FLAT
1	ESPO	0.0323	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
19	ECH	0.0296	iShares MSCI Chile ETF	Chile Equity	LONG
2	EWZ	0.0295	iShares MSCI Brazil ETF	Brazil Equity	FLAT

Cluster 8 – Emerging Markets Equity

Members: 25 **Var explained:** 0.870 **Signal:** LONG since 2026-08-13 (6 days, 0.3%)

Signal logic:

- Latest buy_cross: 2026-08-13 – filter **PASS** (252d at date = +28.6%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +29.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-13 > last sell at 2026-07-07)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7729
Asia Equity	2	0.0815
Asia ex-Japan Equity	1	0.0419
Taiwan Equity	1	0.0362
South Korea Equity	1	0.0339
Global Dividend Equity	1	0.0335

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.036

ticker	rep_wt	name	category	etf_signal
IEMG	0.917	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

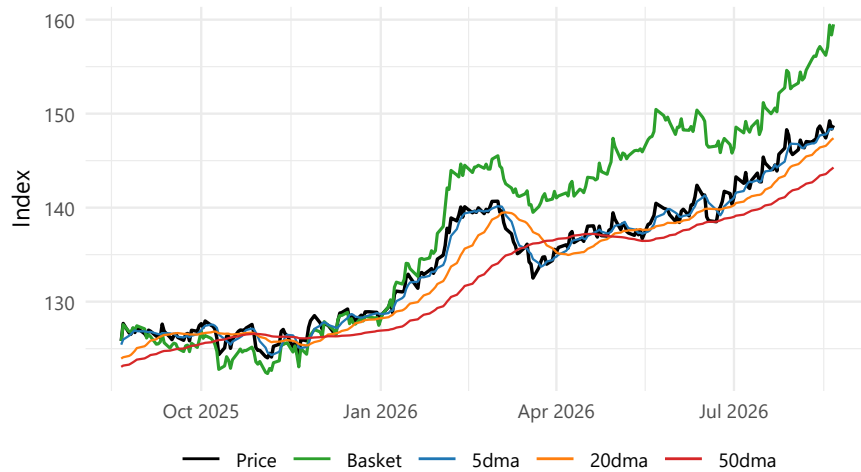
	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0426	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
16	GEM	0.0423	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
20	ESGE	0.0423	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
5	AAXJ	0.0419	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
15	EMGF	0.0419	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
11	EEMA	0.0415	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0412	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
17	FNDE	0.0409	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
12	PXH	0.0408	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
19	EEMV	0.0405	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
10	AIA	0.0403	iShares Asia 50 ETF	Asia Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG
7	DEM	0.0397	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0389	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	LONG
13	DVYE	0.0363	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	LONG
9	EWT	0.0362	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
25	FEMS	0.0350	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	LONG
1	EWY	0.0339	iShares MSCI South Korea ETF	South Korea Equity	FLAT
24	SDIV	0.0335	Global X SuperDividend ETF	Global Dividend Equity	LONG

Cluster 9 – U.S. Dividend Equity

Members: 25 **Var explained:** 0.868 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-20 – filter **PASS** (252d at date = +20.8%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +18.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-20)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	13	0.5283
U.S. Low Volatility Equity	4	0.1562
Value Equity	3	0.1200
Dividend Equity	2	0.0792
U.S. Defensive Equity	1	0.0404
U.S. Quality / Factor Equity	1	0.0390
Global Equity	1	0.0368

Replication basket – SCHD (cor=0.96) basket lead at lag 1: 0.004

ticker	rep_wt	name	category	etf_signal
SCHD	0.891	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
7 SDY	0.0418	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG

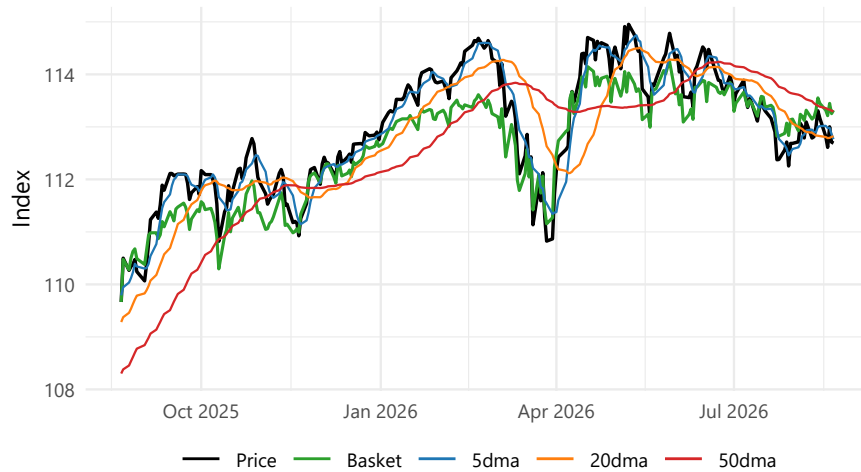
15	DVY	0.0417	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
2	FVD	0.0415	First Trust VL Dividend	U.S. Dividend Equity	LONG
10	SCHD	0.0413	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
14	SDOG	0.0413	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
11	FTA	0.0412	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
20	SPYD	0.0412	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
5	DHS	0.0411	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
19	ONEY	0.0411	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	FLAT
18	SPHD	0.0409	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
6	CDC	0.0404	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
4	FDL	0.0402	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
25	LGLV	0.0401	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
8	LVHD	0.0399	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	FLAT
3	PWV	0.0398	Invesco Large Cap Value ETF	Value Equity	LONG
9	HDV	0.0394	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
16	RDIV	0.0394	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
12	PEY	0.0393	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
17	RPV	0.0391	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
13	DIV	0.0390	Global X Super Dividend ETF	U.S. Dividend Equity	FLAT
21	FTCS	0.0390	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
22	OUSA	0.0383	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
23	SPLV	0.0381	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT
24	USMV	0.0381	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	FLAT
1	ACWV	0.0368	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG

Cluster 10 – High Yield Bond

Members: 21 **Var explained:** 0.764 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +2.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5158
Preferred Stock	7	0.3056
Emerging Markets Fixed Income	4	0.1786

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.046

ticker	rep_wt	name	category	etf_signal
HYG	0.917	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	JNK	0.0527	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
1	HYG	0.0526	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
17	FALN	0.0517	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
6	HYS	0.0512	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0479	iShares Preferred and Income Securities...	Preferred Stock	LONG
3	HYLS	0.0472	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0468	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0468	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
11	PFXF	0.0458	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
20	PCY	0.0458	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0433	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	FLAT
8	VRP	0.0369	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.756 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +34.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.005

ticker	rep_wt	name	category	etf_signal
VHT	1.088	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0656	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0655	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0648	First Trust Health Care AlphaDEX	Healthcare	FLAT
14	RSPH	0.0641	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0640	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0625	iShares Global Healthcare ETF	Healthcare	FLAT

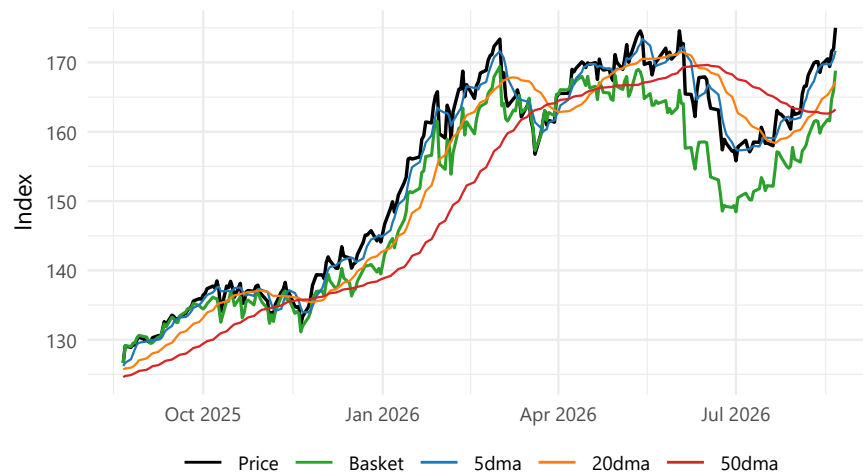
1	XLV	0.0623	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	FLAT
6	FBT	0.0597	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	FLAT
7	XHE	0.0571	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0565	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0546	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0537	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0533	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0507	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0470	iShares U.S. Health Care Providers ETF	Healthcare	FLAT

Cluster 12 – Materials

Members: 17 **Var explained:** 0.682 **Signal:** LONG since 2026-08-05 (12 days, 5.1%)

Signal logic:

- Latest buy_cross: 2026-08-05 – filter **PASS** (252d at date = +34.1%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +39.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-05 > last sell at 2026-06-09)



Category breakdown

category	n_members	abs_weight
Materials	9	0.5600
MLP / Energy Infrastructure	4	0.2164
Energy	1	0.0646
Consumer Staples	1	0.0598
Norway Equity	1	0.0557
Industrial Metals	1	0.0436

Replication basket – AMLP+URA+GUNR (cor=0.98) basket lead at lag 1: 0.047

ticker	rep_wt	name	category	etf_signal
AMLP	0.138	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
URA	0.096	Global X Uranium ETF	Materials	LONG
GUNR	0.772	FlexShares Global Upstream Natural Re...	Materials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

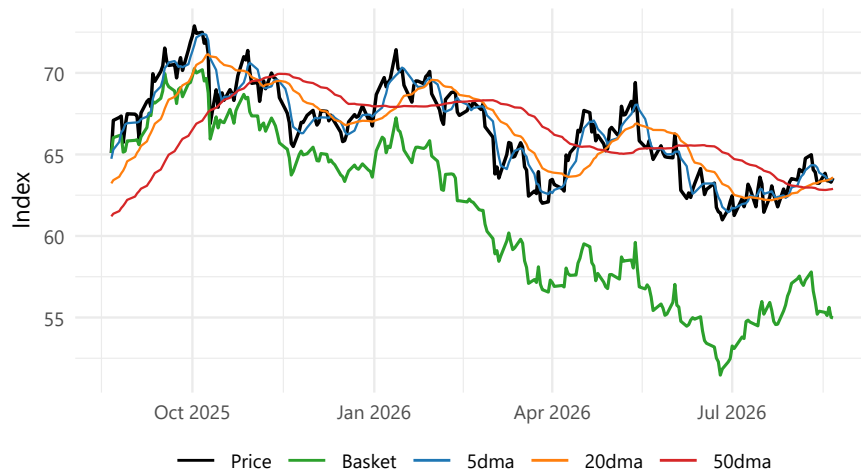
	ticker	PC_wt	name	category	etf_signal
11	GNR	0.0694	State Street SPDR S&P Global Natural Re...	Materials	LONG
10	GUNR	0.0690	FlexShares Global Upstream Natural Reso...	Materials	LONG
17	NANR	0.0670	State Street SPDR S&P North American Na...	Materials	LONG
16	IGE	0.0646	iShares North American Natural Resource...	Energy	LONG
9	MXI	0.0635	iShares Global Materials ETF	Materials	LONG
12	PICK	0.0632	iShares MSCI Global Select Metals & Min...	Materials	LONG
3	XME	0.0613	State Street SPDR S&P Metals & Mining E...	Materials	LONG
13	SLX	0.0608	VanEck Steel ETF	Materials	LONG
5	MOO	0.0598	VanEck Agribusiness ETF	Consumer Staples	FLAT
2	COPX	0.0591	Global X Copper Miners ETF	Materials	LONG
8	MLPX	0.0560	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
14	NORW	0.0557	Global X MSCI Norway ETF	Norway Equity	LONG
4	EMLP	0.0546	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
6	AMLP	0.0536	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
7	MLPA	0.0522	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
1	URA	0.0467	Global X Uranium ETF	Materials	LONG
15	DBB	0.0436	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 13 – China Equity

Members: 15 **Var explained:** 0.794 **Signal:** LONG since 2026-08-03 (14 days, 0.2%)

Signal logic:

- Latest buy_cross: 2026-08-03 – filter **PASS** (252d at date = +2.6%)
- Latest sell_cross: 2026-05-20
- Current 252d return: -2.2% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-08-03 > last sell at 2026-05-20)

**Category breakdown**

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0644
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
KWEB	0.561	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0740	WisdomTree China ex-State-Owned Enterpr...	China Equity	LONG
9	GXC	0.0738	State Street SPDR S&P China ETF	China Equity	LONG
1	MCHI	0.0732	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0668	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0660	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0653	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0644	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0597	KraneShares Boseri MSCI China A 50 Conn...	China Equity	LONG
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
14	CNXT	0.0587	VanEck ChiNext Innovators ETF	China Equity	FLAT
15	KURE	0.0581	KraneShares MSCI All China Health Care ...	China Equity	LONG

Cluster 14 – Financials

Members: 14 **Var explained:** 0.871 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +12.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.9383
Global Equity	1	0.0617

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.012

ticker	rep_wt	name	category	etf_signal
XLF	1.095	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0757	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0756	Fidelity MSCI Financials Index ETF	Financials	LONG
7	IYF	0.0754	iShares U.S. Financial ETF	Financials	FLAT
5	FXO	0.0752	First Trust Financials AlphaDEX	Financials	FLAT
8	IYG	0.0749	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0742	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0733	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0713	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0707	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	FLAT
13	IAI	0.0695	iShares U.S. Broker-Dealers & Securitie...	Financials	FLAT
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0630	State Street SPDR S&P Insurance ETF	Financials	FLAT
12	PSP	0.0617	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-08-06
- Current 252d return: +13.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-06 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.001

ticker	rep_wt	name	category	etf_signal
VNQ	0.968	Vanguard Real Estate ETF	Real Estate	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

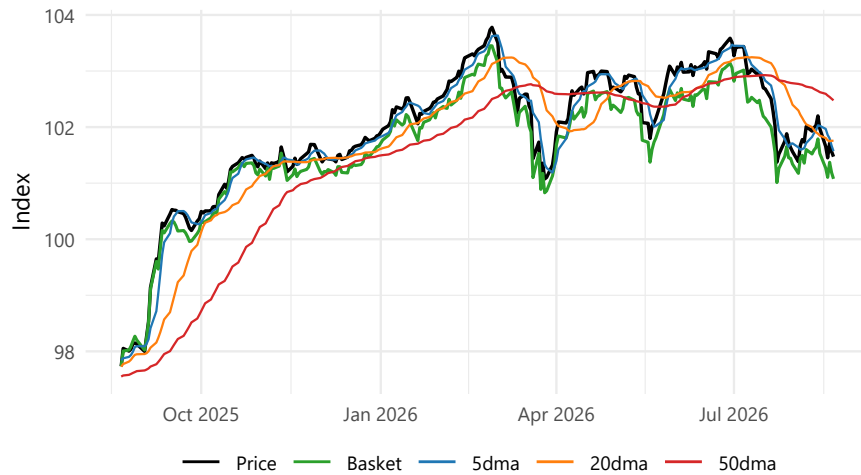
	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	LONG
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	FLAT
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT
1	XLRE	0.0720	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0720	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0716	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0712	State Street SPDR Dow Jones Global Real...	Real Estate	FLAT
4	REZ	0.0696	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0687	Nuveen Short-Term REIT ETF	Real Estate	FLAT
12	INDS	0.0682	Pacer Industrial Real Estate ETF	Real Estate	FLAT

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.820 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-08-21
- Current 252d return: +3.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-21 more recent than last effective buy 2026-05-28)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.178

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0802	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
7	NYF	0.0794	iShares New York Muni Bond ETF	Municipal Bond	FLAT
1	FMB	0.0793	First Trust Managed Municipal ETF	Municipal Bond	FLAT

5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0770	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	FLAT
3	HYMB	0.0722	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	FLAT
12	SUB	0.0693	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – Gold

Members: 12 **Var explained:** 0.864 **Signal:** LONG since 2026-08-07 (10 days, 8.9%)

Signal logic:

- Latest buy_cross: 2026-08-07 – filter **PASS** (252d at date = +36.2%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +47.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-07 > last sell at 2026-05-20)



Category breakdown

category	n_members	abs_weight
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Materials	5	0.4158
Gold	3	0.2503
Precious Metals	2	0.1723
Silver	2	0.1616

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
GLD	0.026	SPDR Gold Shares	Gold	LONG
IAU	0.751	iShares Gold Trust Shares	Gold	LONG
SLV	0.412	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0865	Invesco DB Precious Metals Fund	Precious Metals	LONG
7	GLTR	0.0858	abrdn Physical Precious Metals Basket S...	Precious Metals	LONG
2	GDXJ	0.0841	VanEck Junior Gold Miners ETF	Materials	LONG
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	LONG
3	GLD	0.0835	SPDR Gold Shares	Gold	LONG
9	IAU	0.0835	iShares Gold Trust Shares	Gold	LONG
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	LONG
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	LONG
6	SIL	0.0829	Global X Silver Miners ETF	Materials	LONG
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	LONG
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0807	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.740 **Signal:** LONG since 2026-08-18 (3 days, -0.6%)

Signal logic:

- Latest buy_cross: 2026-08-18 – filter **PASS** (252d at date = +5.0%)
- Latest sell_cross: 2026-08-05
- Current 252d return: +3.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-18 > last sell at 2026-08-05)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5744
Utilities	4	0.3503
U.S. Small Cap Consumer Staples	1	0.0753

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.016

ticker	rep_wt	name	category	etf_signal
XLU	0.363	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.588	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0984	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0984	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0976	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0951	iShares U.S. Consumer Staples ETF	Consumer Staples	FLAT

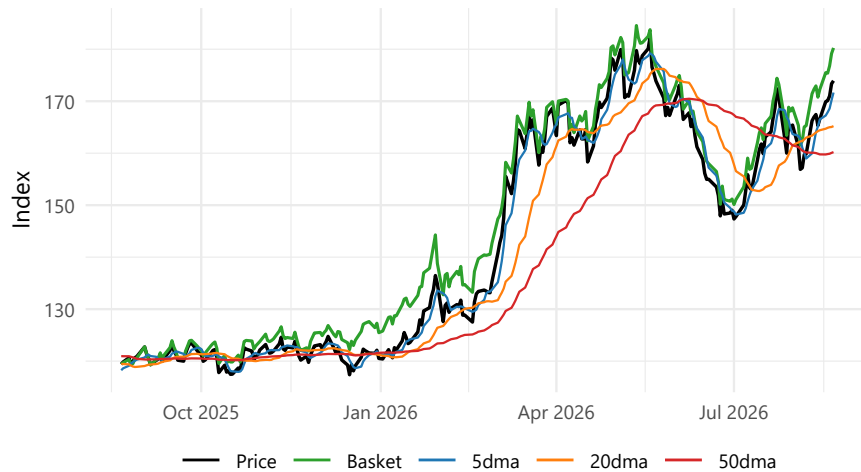
3	KXI	0.0948	iShares Global Consumer Staples ETF	Consumer Staples	FLAT
11	PBJ	0.0900	Invesco Food & Beverage ETF	Consumer Staples	LONG
7	IDU	0.0887	iShares U.S. Utilities ETF	Utilities	FLAT
9	FUTY	0.0874	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
8	VPU	0.0873	Vanguard Utilities ETF	Utilities	FLAT
2	XLU	0.0870	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0753	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 19 – Broad Commodities

Members: 11 **Var explained:** 0.771 **Signal:** LONG since 2026-08-11 (8 days, 4.2%)

Signal logic:

- Latest buy_cross: 2026-08-11 – filter **PASS** (252d at date = +41.2%)
- Latest sell_cross: 2026-08-06
- Current 252d return: +46.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-11 > last sell at 2026-08-06)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6174
Oil	2	0.1980
Energy Commodities	1	0.1019
Agricultural Commodities	1	0.0495
Natural Gas	1	0.0332

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.079

ticker	rep_wt	name	category	etf_signal
PDBC	1.125	Invesco Optimum Yield Diversified Com...	Broad Commodities	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1056	Invesco DB Commodity Index Tracking Fund	Broad Commodities	FLAT
6	GSG	0.1056	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
8	PDBC	0.1056	Invesco Optimum Yield Diversified Commo...	Broad Commodities	FLAT
9	COMT	0.1049	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
10	DBE	0.1019	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.0996	United States Oil Fund	Oil	LONG
5	DBO	0.0984	Invesco DB Oil Fund	Oil	LONG
7	DJP	0.0984	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
4	FTGC	0.0972	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
3	DBA	0.0495	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
11	UNG	0.0332	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 20 – India Equity

Members: 10 **Var explained:** 0.405 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-08-03 – filter **BLOCKED** (252d at date = -10.3%; needs > 0)
- Latest effective buy: 2026-05-06 – delayed re-arm from blocked cross (252d at fire = +4.7%)
- Latest sell_cross: 2026-07-27
- Current 252d return: -13.2% – filter **BLOCKS** new entries today

- Decision: **FLAT** (last sell 2026-07-27 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4677
Malaysia Equity	1	0.1126
Indonesia Equity	1	0.1037
Philippines Equity	1	0.1013
Vietnam Equity	1	0.0751
Turkey Equity	1	0.0608
U.S. Dividend Equity	1	0.0548
Platinum	1	0.0240

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.005

ticker	rep_wt	name	category	etf_signal
NOBL	0.08	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.66	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.03	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1573	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1556	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1549	WisdomTree India Earnings Fund	India Equity	FLAT
8	EWM	0.1126	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
6	EIDO	0.1037	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1013	iShares MSCI Philippines ETF	Philippines Equity	FLAT
10	VNM	0.0751	VanEck Vietnam ETF	Vietnam Equity	FLAT
9	TUR	0.0608	iShares MSCI Turkey ETF	Turkey Equity	LONG
4	NOBL	0.0548	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0240	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.904 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-07-02
- Current 252d return: -0.6% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VTIP	0.267	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.760	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1036	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	FLAT
6	TIPX	0.1035	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1024	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1008	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.1000	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT

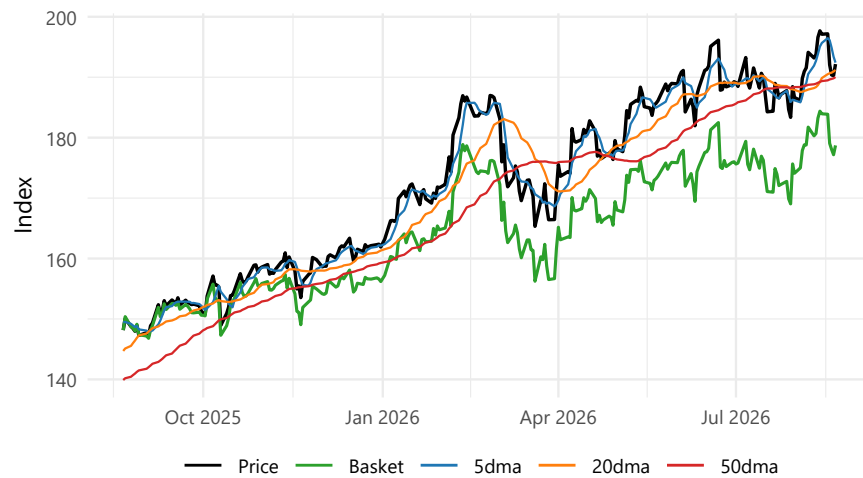
9	STIP	0.0997	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0989	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0884	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.887 **Signal:** LONG since 2026-08-06 (11 days, 0.3%)

Signal logic:

- Latest buy_cross: 2026-08-06 – filter **PASS** (252d at date = +35.6%)
- Latest sell_cross: 2026-07-28
- Current 252d return: +28.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-06 > last sell at 2026-07-28)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7805

Asia-Pacific Equity	2	0.2195
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Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.043

ticker	rep_wt	name	category	etf_signal
EWJ	0.907	iShares MSCI Japan Index Fund	Japan Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

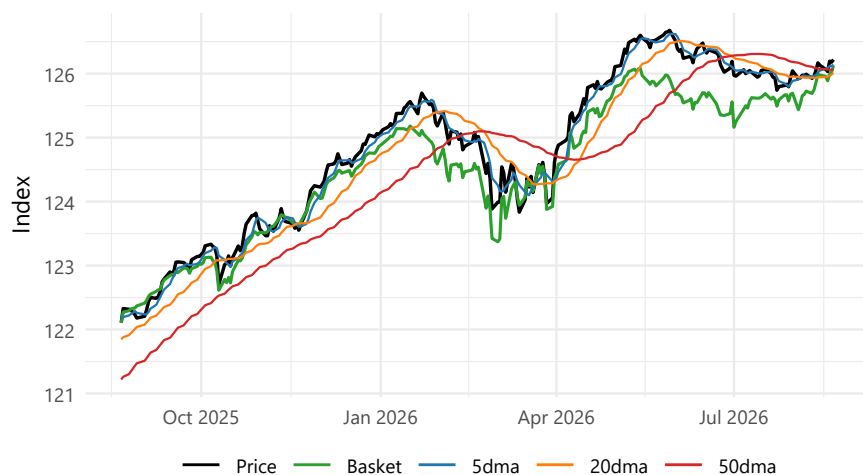
	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1142	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
6	DBJP	0.1118	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
9	HEWJ	0.1118	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
3	IPAC	0.1114	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	LONG
5	FLJH	0.1110	Franklin FTSE Japan Hedged ETF	Japan Equity	LONG
2	DXJ	0.1095	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1081	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1079	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.555 **Signal:** LONG since 2026-08-17 (4 days, 0.1%)

Signal logic:

- Latest buy_cross: 2026-08-17 – filter **PASS** (252d at date = +3.2%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +3.3% – filter **ALLOWES** new entries today
- Decision: **LONG** (effective buy at 2026-08-17 > last sell at 2026-07-23)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3619
Interest Rate Hedged Fixed Income	3	0.3245
Floating Rate Fixed Income	3	0.3136

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.293	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	LONG
BKLN	0.217	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.299	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.123	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 SRLN	0.1251	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG

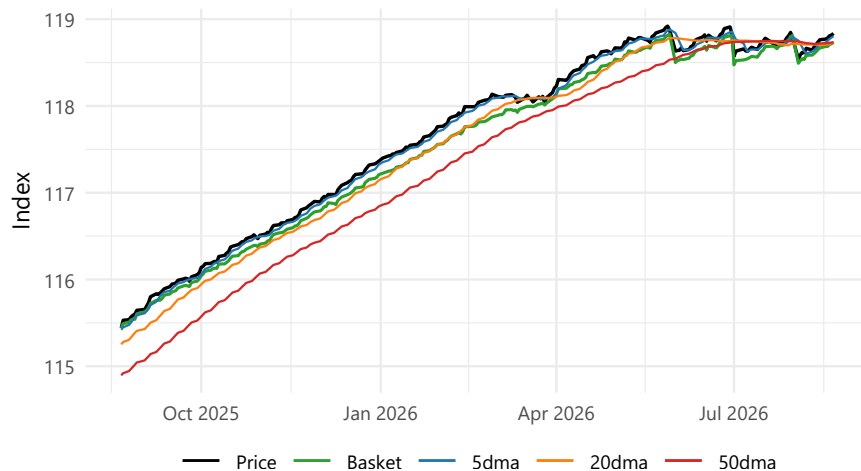
3	BKLN	0.1208	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1160	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1143	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1085	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	LONG
6	IGBH	0.1056	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1047	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1039	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1012	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	LONG

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 **Var explained:** 0.461 **Signal:** LONG since 2026-08-17 (4 days, 0.1%)

Signal logic:

- Latest buy_cross: 2026-08-17 – filter **PASS** (252d at date = +3.0%)
- Latest sell_cross: 2026-08-05
- Current 252d return: +2.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-17 > last sell at 2026-08-05)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.86) basket lead at lag 1: 0.047

ticker	rep_wt	name	category	etf_signal
SHV	0.255	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	LONG
MINT	0.173	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.125	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.399	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1325	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	LONG
6	ICSH	0.1307	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	LONG
2	MINT	0.1156	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
5	GSY	0.1124	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
4	NEAR	0.1101	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.1065	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1056	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	LONG
9	GBIL	0.0948	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.0918	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-08-04 – filter **BLOCKED** (252d at date = -1.4%; needs > 0)
- Latest effective buy: 2026-05-29 – delayed re-arm from blocked cross (252d at fire = +0.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: -1.2% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-29)



Category breakdown

category	n_members	abs_weight
Currency	4	0.6409
International Fixed Income	2	0.3591

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.023

ticker	rep_wt	name	category	etf_signal
BWX	0.752	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	LONG
5	FXF	0.1651	Invesco CurrencyShares Swiss Franc Trust	Currency	LONG
3	FXB	0.1608	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1448	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.625 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-07-20 – filter **BLOCKED** (252d at date = -31.3%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-08-05
- Current 252d return: -34.6% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-08-05 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.022

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT
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Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
VHT	cluster_11	Healthcare	2026-05-19	65	19.6%	0.00	1.00
IYH	cluster_11	Healthcare	2026-05-20	64	19.4%	0.00	1.00
FHLC	cluster_11	Healthcare	2026-05-19	65	19.5%	0.00	1.00
XLV	cluster_11	Healthcare	2026-05-21	63	17.9%	0.00	1.00
PTH	cluster_11	Healthcare	2026-06-17	45	19.3%	0.00	1.00
PPH	cluster_11	Healthcare	2026-05-21	63	12.4%	0.00	1.00
GDX	cluster_17	Gold	2026-08-06	11	22.5%	0.00	1.00
IAU	cluster_17	Gold	2026-08-07	10	6.3%	0.00	1.00
DBP	cluster_17	Gold	2026-08-07	10	6.9%	0.00	1.00
GLD	cluster_17	Gold	2026-08-07	10	6.2%	0.00	1.00
SGOL	cluster_17	Gold	2026-08-07	10	6.3%	0.00	1.00
GDXJ	cluster_17	Gold	2026-08-06	11	21.2%	0.00	1.00
SIL	cluster_17	Gold	2026-08-07	10	12.5%	0.00	1.00
GOAU	cluster_17	Gold	2026-08-06	11	19.9%	0.00	1.00
GLTR	cluster_17	Gold	2026-08-10	9	5.1%	0.00	1.00
SLV	cluster_17	Gold	2026-08-10	9	5.6%	0.00	1.00
SIVR	cluster_17	Gold	2026-08-10	9	5.5%	0.00	1.00
SILJ	cluster_17	Gold	2026-08-07	10	9.6%	0.00	1.00
NANR	cluster_12	Materials	2026-07-24	20	12.8%	0.00	1.00
NORW	cluster_12	Materials	2026-08-05	12	6.8%	0.00	1.00
IGE	cluster_12	Materials	2026-07-23	21	10.2%	0.00	1.00
GNR	cluster_12	Materials	2026-07-27	19	11.2%	0.00	1.00
GUNR	cluster_12	Materials	2026-07-27	19	9.8%	0.00	1.00
MXI	cluster_12	Materials	2026-08-05	12	4.4%	0.00	1.00
XME	cluster_12	Materials	2026-08-10	9	1.5%	0.00	1.00
COPX	cluster_12	Materials	2026-08-05	12	9.1%	0.00	1.00
URA	cluster_12	Materials	2026-08-11	8	2.0%	0.01	0.99
PICK	cluster_12	Materials	2026-08-05	12	3.9%	0.01	0.99
EMLP	cluster_12	Materials	2026-08-13	6	-1.0%	0.14	0.78
MLPA	cluster_12	Materials	2026-07-10	30	4.2%	0.55	0.45
MLPX	cluster_12	Materials	2026-08-17	4	-0.9%	0.24	0.43
DBB	cluster_12	Materials	2026-08-04	13	1.0%	0.63	0.37
SLX	cluster_12	Materials	2026-07-30	16	0.7%	0.82	0.18
AMPLP	cluster_12	Materials	2026-07-10	30	3.9%	0.91	0.09
BKLN	cluster_23	Bank Loan / Floating Ra...	2026-08-05	12	0.6%	0.00	1.00
SRLN	cluster_23	Bank Loan / Floating Ra...	2026-07-30	16	0.6%	0.00	1.00
FTSL	cluster_23	Bank Loan / Floating Ra...	2026-08-07	10	-0.3%	0.02	0.98

FLTR	cluster_23	Bank Loan / Floating Ra...	2026-08-21	0	0.0%	0.32	0.19
FLOT	cluster_23	Bank Loan / Floating Ra...	2026-08-21	0	0.0%	0.41	0.04
FXE	cluster_25	International Fixed Inc...	2026-08-04	13	1.3%	0.00	1.00
FXB	cluster_25	International Fixed Inc...	2026-07-31	15	1.2%	0.00	1.00
FXF	cluster_25	International Fixed Inc...	2026-08-20	1	-0.0%	0.00	0.99
SCHC	cluster_4	Developed ex-US Equity	2026-08-05	12	2.3%	0.00	1.00
FNDF	cluster_4	Developed ex-US Equity	2026-07-31	15	3.4%	0.00	1.00
GWX	cluster_4	Developed ex-US Equity	2026-08-06	11	2.2%	0.00	1.00
VIDI	cluster_4	Developed ex-US Equity	2026-08-10	9	1.7%	0.00	1.00
VSS	cluster_4	Developed ex-US Equity	2026-08-07	10	0.9%	0.00	1.00
IQLT	cluster_4	Developed ex-US Equity	2026-07-01	36	6.3%	0.00	1.00
DLS	cluster_4	Developed ex-US Equity	2026-08-03	14	2.8%	0.00	1.00
PXF	cluster_4	Developed ex-US Equity	2026-08-03	14	3.3%	0.00	1.00
FEP	cluster_4	Developed ex-US Equity	2026-07-31	15	2.5%	0.00	1.00
SPDW	cluster_4	Developed ex-US Equity	2026-08-05	12	1.5%	0.01	0.99
VEA	cluster_4	Developed ex-US Equity	2026-08-05	12	1.5%	0.01	0.99
SCHF	cluster_4	Developed ex-US Equity	2026-08-05	12	1.5%	0.01	0.99
IQDF	cluster_4	Developed ex-US Equity	2026-08-04	13	1.5%	0.01	0.99
TLTD	cluster_4	Developed ex-US Equity	2026-07-30	16	3.1%	0.01	0.99
VXUS	cluster_4	Developed ex-US Equity	2026-08-05	12	1.4%	0.01	0.99
ACWX	cluster_4	Developed ex-US Equity	2026-08-05	12	1.3%	0.01	0.99
IXUS	cluster_4	Developed ex-US Equity	2026-08-05	12	1.3%	0.01	0.99
VEU	cluster_4	Developed ex-US Equity	2026-08-05	12	1.3%	0.01	0.99
IMTM	cluster_4	Developed ex-US Equity	2026-08-05	12	1.5%	0.01	0.98
FNDC	cluster_4	Developed ex-US Equity	2026-08-04	13	1.2%	0.02	0.98
CWI	cluster_4	Developed ex-US Equity	2026-08-04	13	1.0%	0.02	0.98
IDV	cluster_4	Developed ex-US Equity	2026-07-28	18	3.6%	0.02	0.98
INTF	cluster_4	Developed ex-US Equity	2026-07-27	19	4.1%	0.03	0.97
EFG	cluster_4	Developed ex-US Equity	2026-08-05	12	0.8%	0.03	0.96
EWG	cluster_4	Developed ex-US Equity	2026-07-31	15	3.2%	0.04	0.96
QEFA	cluster_4	Developed ex-US Equity	2026-07-06	34	4.2%	0.04	0.96
HFXI	cluster_4	Developed ex-US Equity	2026-08-05	12	0.9%	0.03	0.96
SCZ	cluster_4	Developed ex-US Equity	2026-08-04	13	1.4%	0.05	0.95
EZU	cluster_4	Developed ex-US Equity	2026-07-31	15	2.8%	0.05	0.95
FEZ	cluster_4	Developed ex-US Equity	2026-07-30	16	3.0%	0.06	0.94
JPIN	cluster_4	Developed ex-US Equity	2026-07-27	19	4.0%	0.06	0.94
IEFA	cluster_4	Developed ex-US Equity	2026-07-30	16	2.0%	0.06	0.94
GSIE	cluster_4	Developed ex-US Equity	2026-07-02	35	4.8%	0.07	0.93
EWN	cluster_4	Developed ex-US Equity	2026-08-05	12	1.0%	0.03	0.93
VGK	cluster_4	Developed ex-US Equity	2026-07-01	36	5.6%	0.07	0.93
IEUR	cluster_4	Developed ex-US Equity	2026-07-30	16	1.7%	0.08	0.92
EFA	cluster_4	Developed ex-US Equity	2026-07-30	16	1.9%	0.08	0.92
RODM	cluster_4	Developed ex-US Equity	2026-07-27	19	2.3%	0.09	0.91
EPOL	cluster_4	Developed ex-US Equity	2026-07-08	32	11.7%	0.09	0.91
IEV	cluster_4	Developed ex-US Equity	2026-07-27	19	4.1%	0.09	0.91

ESGD	cluster_4	Developed ex-US Equity	2026-07-30	16	1.9%	0.11	0.89
IVLU	cluster_4	Developed ex-US Equity	2026-07-27	19	3.9%	0.13	0.87
DWM	cluster_4	Developed ex-US Equity	2026-07-30	16	1.7%	0.17	0.83
EW0	cluster_4	Developed ex-US Equity	2026-07-31	15	2.8%	0.20	0.77
HEZU	cluster_4	Developed ex-US Equity	2026-07-30	16	1.6%	0.21	0.76
EWU	cluster_4	Developed ex-US Equity	2026-07-16	26	4.2%	0.28	0.72
VYMI	cluster_4	Developed ex-US Equity	2026-07-08	32	5.7%	0.29	0.71
EFV	cluster_4	Developed ex-US Equity	2026-07-08	32	5.9%	0.30	0.70
EWD	cluster_4	Developed ex-US Equity	2026-07-30	16	2.2%	0.35	0.65
EUFN	cluster_4	Developed ex-US Equity	2026-06-12	48	8.7%	0.35	0.65
HDEF	cluster_4	Developed ex-US Equity	2026-07-09	31	5.2%	0.36	0.64
EWQ	cluster_4	Developed ex-US Equity	2026-07-30	16	1.1%	0.47	0.53
VIGI	cluster_4	Developed ex-US Equity	2026-06-30	37	5.6%	0.49	0.51
IPKW	cluster_4	Developed ex-US Equity	2026-07-10	30	5.2%	0.52	0.48
HEDJ	cluster_4	Developed ex-US Equity	2026-08-03	14	-0.5%	0.68	0.11
USO	cluster_19	Broad Commodities	2026-08-11	8	5.5%	0.17	0.83
DBO	cluster_19	Broad Commodities	2026-08-11	8	5.4%	0.21	0.79
SCHD	cluster_9	U.S. Dividend Equity	2026-07-07	33	7.9%	0.00	1.00
PWV	cluster_9	U.S. Dividend Equity	2026-04-02	97	17.8%	0.00	1.00
HDV	cluster_9	U.S. Dividend Equity	2026-06-25	40	8.3%	0.00	1.00
FDL	cluster_9	U.S. Dividend Equity	2026-07-16	26	3.6%	0.01	0.99
FTA	cluster_9	U.S. Dividend Equity	2026-05-21	63	10.2%	0.03	0.97
FTCS	cluster_9	U.S. Dividend Equity	2026-06-26	39	8.0%	0.05	0.95
ACWV	cluster_9	U.S. Dividend Equity	2026-07-06	34	4.1%	0.06	0.94
SPHD	cluster_9	U.S. Dividend Equity	2026-06-08	52	6.6%	0.13	0.87
SDY	cluster_9	U.S. Dividend Equity	2026-05-20	64	7.0%	0.15	0.85
OUA	cluster_9	U.S. Dividend Equity	2026-07-01	36	5.7%	0.29	0.71
FVD	cluster_9	U.S. Dividend Equity	2026-06-10	50	6.2%	0.52	0.48
LGLV	cluster_9	U.S. Dividend Equity	2026-06-26	39	3.3%	0.76	0.24
EEMV	cluster_8	Emerging Markets Equity	2026-08-12	7	0.1%	0.00	1.00
FEM	cluster_8	Emerging Markets Equity	2026-08-07	10	2.1%	0.00	1.00
DVYE	cluster_8	Emerging Markets Equity	2026-07-24	20	4.9%	0.02	0.98
FEMS	cluster_8	Emerging Markets Equity	2026-08-12	7	1.5%	0.00	0.98
EEMA	cluster_8	Emerging Markets Equity	2026-08-12	7	0.2%	0.01	0.97
EMGF	cluster_8	Emerging Markets Equity	2026-08-14	5	0.4%	0.00	0.95
SPEM	cluster_8	Emerging Markets Equity	2026-08-06	11	0.9%	0.04	0.95
DEM	cluster_8	Emerging Markets Equity	2026-08-07	10	0.7%	0.05	0.94
VWO	cluster_8	Emerging Markets Equity	2026-08-06	11	0.8%	0.05	0.94
ESGE	cluster_8	Emerging Markets Equity	2026-08-14	5	0.7%	0.00	0.94
IEMG	cluster_8	Emerging Markets Equity	2026-08-13	6	0.3%	0.00	0.93
SCHE	cluster_8	Emerging Markets Equity	2026-08-05	12	0.8%	0.06	0.93
EWX	cluster_8	Emerging Markets Equity	2026-08-14	5	0.4%	0.00	0.93
EEM	cluster_8	Emerging Markets Equity	2026-08-14	5	0.8%	0.00	0.91
GEM	cluster_8	Emerging Markets Equity	2026-08-13	6	0.3%	0.01	0.91
AAXJ	cluster_8	Emerging Markets Equity	2026-08-14	5	0.4%	0.00	0.90

EWT	cluster_8	Emerging Markets Equity	2026-08-11	8	0.3%	0.01	0.90
GMF	cluster_8	Emerging Markets Equity	2026-08-06	11	0.2%	0.07	0.88
XSOE	cluster_8	Emerging Markets Equity	2026-08-17	4	-0.9%	0.01	0.85
AIA	cluster_8	Emerging Markets Equity	2026-08-14	5	-0.1%	0.02	0.83
DGS	cluster_8	Emerging Markets Equity	2026-08-12	7	-1.2%	0.05	0.64
FNDE	cluster_8	Emerging Markets Equity	2026-07-30	16	2.1%	0.37	0.63
PXH	cluster_8	Emerging Markets Equity	2026-07-31	15	1.3%	0.46	0.54
SDIV	cluster_8	Emerging Markets Equity	2026-08-19	2	0.0%	0.64	0.29
GSY	cluster_24	Cash / Ultra-Short Fixe...	2026-08-07	10	0.2%	0.00	1.00
NEAR	cluster_24	Cash / Ultra-Short Fixe...	2026-08-18	3	0.0%	0.00	0.99
FTSM	cluster_24	Cash / Ultra-Short Fixe...	2026-08-17	4	0.1%	0.08	0.82
ICSH	cluster_24	Cash / Ultra-Short Fixe...	2026-08-17	4	0.0%	0.15	0.75
MINT	cluster_24	Cash / Ultra-Short Fixe...	2026-08-17	4	0.1%	0.23	0.59
GBIL	cluster_24	Cash / Ultra-Short Fixe...	2026-08-18	3	0.0%	0.24	0.57
SHV	cluster_24	Cash / Ultra-Short Fixe...	2026-08-18	3	0.0%	0.28	0.51
VNLA	cluster_24	Cash / Ultra-Short Fixe...	2026-08-21	0	0.0%	0.15	0.45
BIL	cluster_24	Cash / Ultra-Short Fixe...	2026-08-18	3	0.0%	0.31	0.41
IYM	cluster_3	U.S. Large Cap Value	2026-08-06	11	5.8%	0.00	1.00
VLUE	cluster_3	U.S. Large Cap Value	2026-08-07	10	2.5%	0.00	1.00
VTV	cluster_3	U.S. Large Cap Value	2026-04-13	91	11.7%	0.00	1.00
FNDX	cluster_3	U.S. Large Cap Value	2026-04-09	93	13.0%	0.01	0.99
SCHV	cluster_3	U.S. Large Cap Value	2026-08-04	13	0.1%	0.01	0.99
PRF	cluster_3	U.S. Large Cap Value	2026-04-09	93	13.4%	0.02	0.98
RSP	cluster_3	U.S. Large Cap Value	2026-04-14	90	11.0%	0.02	0.98
FEX	cluster_3	U.S. Large Cap Value	2026-04-09	93	10.3%	0.03	0.97
DTD	cluster_3	U.S. Large Cap Value	2026-04-13	91	9.5%	0.03	0.97
DLN	cluster_3	U.S. Large Cap Value	2026-04-13	91	9.6%	0.04	0.96
PKW	cluster_3	U.S. Large Cap Value	2026-05-20	64	11.0%	0.05	0.95
VAW	cluster_3	U.S. Large Cap Value	2026-08-06	11	2.9%	0.05	0.94
JVAL	cluster_3	U.S. Large Cap Value	2026-07-30	16	3.4%	0.06	0.93
VONV	cluster_3	U.S. Large Cap Value	2026-04-10	92	15.9%	0.08	0.92
IWD	cluster_3	U.S. Large Cap Value	2026-04-10	92	16.0%	0.08	0.92
DGRO	cluster_3	U.S. Large Cap Value	2026-04-13	91	10.1%	0.09	0.91
IWX	cluster_3	U.S. Large Cap Value	2026-04-10	92	16.9%	0.13	0.87
IYZ	cluster_3	U.S. Large Cap Value	2026-08-06	11	-0.8%	0.13	0.87
RWL	cluster_3	U.S. Large Cap Value	2026-04-10	92	13.1%	0.18	0.82
XLB	cluster_3	U.S. Large Cap Value	2026-07-28	18	2.3%	0.19	0.81
RSPM	cluster_3	U.S. Large Cap Value	2026-07-27	19	4.8%	0.20	0.80
VIG	cluster_3	U.S. Large Cap Value	2026-04-14	90	9.0%	0.25	0.75
FIDU	cluster_3	U.S. Large Cap Value	2026-08-05	12	-3.3%	0.63	-0.21
VIS	cluster_3	U.S. Large Cap Value	2026-08-05	12	-3.4%	0.66	-0.29
ITB	cluster_3	U.S. Large Cap Value	2026-07-28	18	-2.8%	0.52	-0.35
XHB	cluster_3	U.S. Large Cap Value	2026-08-07	10	-3.9%	0.66	-0.62
AIVL	cluster_3	U.S. Large Cap Value	2026-04-10	92	11.4%		
FV	cluster_1	U.S. Large Cap Equity	2026-08-10	9	1.7%	0.00	1.00

QQEW	cluster_1	U.S. Large Cap Equity	2026-08-04	13	1.3%	0.00	1.00
QAI	cluster_1	U.S. Large Cap Equity	2026-08-06	11	0.7%	0.02	0.98
IWP	cluster_1	U.S. Large Cap Equity	2026-08-07	10	-0.3%	0.02	0.97
XT	cluster_1	U.S. Large Cap Equity	2026-08-07	10	0.1%	0.03	0.96
AOM	cluster_1	U.S. Large Cap Equity	2026-08-06	11	0.1%	0.04	0.95
SPGM	cluster_1	U.S. Large Cap Equity	2026-08-04	13	0.3%	0.05	0.95
VT	cluster_1	U.S. Large Cap Equity	2026-08-04	13	0.2%	0.05	0.95
AOA	cluster_1	U.S. Large Cap Equity	2026-08-05	12	-0.0%	0.05	0.94
AOR	cluster_1	U.S. Large Cap Equity	2026-08-05	12	-0.0%	0.06	0.94
SUSA	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.2%	0.07	0.93
ACWI	cluster_1	U.S. Large Cap Equity	2026-08-04	13	0.1%	0.07	0.93
CRBN	cluster_1	U.S. Large Cap Equity	2026-08-04	13	0.0%	0.07	0.93
DSI	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.3%	0.07	0.93
VOT	cluster_1	U.S. Large Cap Equity	2026-08-05	12	-0.0%	0.06	0.92
EIS	cluster_1	U.S. Large Cap Equity	2026-08-12	7	-0.3%	0.08	0.91
URTH	cluster_1	U.S. Large Cap Equity	2026-08-03	14	1.6%	0.10	0.90
VONE	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.6%	0.11	0.89
AOK	cluster_1	U.S. Large Cap Equity	2026-08-06	11	0.0%	0.07	0.89
IWV	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.6%	0.11	0.89
VTHR	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.6%	0.12	0.88
IWB	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.6%	0.12	0.88
EPS	cluster_1	U.S. Large Cap Equity	2026-08-03	14	0.9%	0.12	0.88
VTI	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.7%	0.12	0.88
ILCB	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.6%	0.12	0.88
IYY	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.6%	0.12	0.87
ITOT	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.7%	0.12	0.87
JHML	cluster_1	U.S. Large Cap Equity	2026-08-03	14	1.0%	0.13	0.87
SCHB	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.6%	0.13	0.87
DGRW	cluster_1	U.S. Large Cap Equity	2026-08-03	14	1.5%	0.13	0.87
VV	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.7%	0.13	0.87
GSLC	cluster_1	U.S. Large Cap Equity	2026-08-03	14	1.1%	0.14	0.86
LRGF	cluster_1	U.S. Large Cap Equity	2026-08-03	14	1.2%	0.14	0.86
SPY	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.7%	0.14	0.85
IVV	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.7%	0.14	0.85
VOO	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.7%	0.14	0.85
SCHX	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.7%	0.15	0.85
SPTM	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.8%	0.15	0.85
IWL	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.9%	0.16	0.84
MGC	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.9%	0.16	0.84
ESGU	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.8%	0.16	0.83
CATH	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.9%	0.17	0.83
PRNT	cluster_1	U.S. Large Cap Equity	2026-08-05	12	-1.8%	0.18	0.82
IMCG	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.5%	0.15	0.81
OEF	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-1.0%	0.19	0.80
SPYX	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-1.0%	0.22	0.78

IYC	cluster_1	U.S. Large Cap Equity	2026-08-04	13	-0.2%	0.21	0.78
XLC	cluster_1	U.S. Large Cap Equity	2026-08-06	11	0.2%	0.21	0.78
SPHB	cluster_1	U.S. Large Cap Equity	2026-08-12	7	-1.0%	0.04	0.74
XLY	cluster_1	U.S. Large Cap Equity	2026-08-05	12	-0.5%	0.21	0.72
IOO	cluster_1	U.S. Large Cap Equity	2026-08-03	14	0.5%	0.28	0.72
XAR	cluster_1	U.S. Large Cap Equity	2026-08-06	11	-3.6%	0.23	0.72
XLG	cluster_1	U.S. Large Cap Equity	2026-08-05	12	-1.4%	0.25	0.72
BOTZ	cluster_1	U.S. Large Cap Equity	2026-08-07	10	-4.2%	0.23	0.67
VFMO	cluster_1	U.S. Large Cap Equity	2026-08-13	6	-1.7%	0.03	0.60
FDIS	cluster_1	U.S. Large Cap Equity	2026-08-05	12	-0.9%	0.28	0.60
VCR	cluster_1	U.S. Large Cap Equity	2026-08-05	12	-1.0%	0.28	0.59
ROBO	cluster_1	U.S. Large Cap Equity	2026-08-07	10	-3.9%	0.29	0.51
CWB	cluster_1	U.S. Large Cap Equity	2026-08-14	5	-2.7%	0.11	0.46
RPG	cluster_1	U.S. Large Cap Equity	2026-08-17	4	-4.5%	0.04	0.45
FTC	cluster_1	U.S. Large Cap Equity	2026-08-10	9	-1.4%	0.13	0.41
SPHQ	cluster_1	U.S. Large Cap Equity	2026-08-14	5	-1.9%	0.04	0.39
ICVT	cluster_1	U.S. Large Cap Equity	2026-08-17	4	-2.7%	0.08	0.38
QMOM	cluster_1	U.S. Large Cap Equity	2026-08-12	7	-3.5%	0.21	0.37
ITA	cluster_1	U.S. Large Cap Equity	2026-07-24	20	-1.2%	0.63	0.25
PDP	cluster_1	U.S. Large Cap Equity	2026-08-17	4	-5.7%	0.08	0.22
PPA	cluster_1	U.S. Large Cap Equity	2026-07-27	19	-2.0%	0.66	0.21
FCOM	cluster_1	U.S. Large Cap Equity	2026-08-13	6	-1.4%	0.45	0.07
MTUM	cluster_1	U.S. Large Cap Equity	2026-08-18	3	-2.1%	0.12	0.07
VOX	cluster_1	U.S. Large Cap Equity	2026-08-13	6	-1.4%	0.45	0.04
XMMO	cluster_1	U.S. Large Cap Equity	2026-08-17	4	-5.6%	0.24	0.03
DWAS	cluster_1	U.S. Large Cap Equity	2026-08-14	5	-4.3%	0.29	-0.16
FXC	cluster_7	Asia-Pacific Equity	2026-08-19	2	0.3%	0.00	1.00
FXA	cluster_7	Asia-Pacific Equity	2026-08-04	13	1.9%	0.00	1.00
EMLC	cluster_7	Asia-Pacific Equity	2026-08-05	12	1.0%	0.00	1.00
EWC	cluster_7	Asia-Pacific Equity	2026-07-10	30	6.3%	0.00	1.00
EZA	cluster_7	Asia-Pacific Equity	2026-08-05	12	8.5%	0.00	1.00
EBND	cluster_7	Asia-Pacific Equity	2026-07-31	15	1.4%	0.00	1.00
EWS	cluster_7	Asia-Pacific Equity	2026-06-17	45	14.1%	0.00	1.00
EWK	cluster_7	Asia-Pacific Equity	2026-08-18	3	2.2%	0.01	0.99
ENZL	cluster_7	Asia-Pacific Equity	2026-07-07	33	7.2%	0.05	0.95
NFRA	cluster_7	Asia-Pacific Equity	2026-07-16	26	1.1%	0.07	0.93
ECH	cluster_7	Asia-Pacific Equity	2026-08-05	12	1.1%	0.06	0.93
EFAV	cluster_7	Asia-Pacific Equity	2026-07-16	26	4.8%	0.21	0.79
EWL	cluster_7	Asia-Pacific Equity	2026-07-28	18	1.8%	0.29	0.70
EPP	cluster_7	Asia-Pacific Equity	2026-07-15	27	4.7%	0.33	0.67
EWA	cluster_7	Asia-Pacific Equity	2026-07-20	24	5.7%	0.50	0.50
BBAX	cluster_7	Asia-Pacific Equity	2026-07-14	28	5.3%	0.50	0.50
DWX	cluster_7	Asia-Pacific Equity	2026-07-09	31	3.4%	0.74	0.26
IDLV	cluster_7	Asia-Pacific Equity	2026-07-10	30	3.1%	0.91	0.09
ARKK	cluster_6	U.S. Large Cap Growth	2026-08-10	9	7.2%	0.00	1.00

ARKW	cluster_6	U.S. Large Cap Growth	2026-08-21	0	0.0%	0.00	1.00
SKYY	cluster_6	U.S. Large Cap Growth	2026-07-30	16	15.1%	0.01	0.99
QYLD	cluster_6	U.S. Large Cap Growth	2026-08-07	10	1.0%	0.00	0.99
FXL	cluster_6	U.S. Large Cap Growth	2026-08-06	11	1.3%	0.03	0.97
ARKQ	cluster_6	U.S. Large Cap Growth	2026-08-11	8	-2.1%	0.02	0.95
AIQ	cluster_6	U.S. Large Cap Growth	2026-08-10	9	-0.1%	0.03	0.92
NXTG	cluster_6	U.S. Large Cap Growth	2026-08-12	7	-2.3%	0.05	0.90
PTLC	cluster_6	U.S. Large Cap Growth	2026-08-04	13	-0.7%	0.15	0.85
SCHG	cluster_6	U.S. Large Cap Growth	2026-08-04	13	-0.6%	0.17	0.83
VGT	cluster_6	U.S. Large Cap Growth	2026-08-06	11	-1.0%	0.13	0.83
FTEC	cluster_6	U.S. Large Cap Growth	2026-08-06	11	-1.0%	0.13	0.82
FDN	cluster_6	U.S. Large Cap Growth	2026-08-03	14	1.4%	0.19	0.81
ONEQ	cluster_6	U.S. Large Cap Growth	2026-08-06	11	-0.7%	0.15	0.80
IXN	cluster_6	U.S. Large Cap Growth	2026-08-07	10	-1.0%	0.07	0.80
TDIV	cluster_6	U.S. Large Cap Growth	2026-08-07	10	-2.1%	0.18	0.80
VONG	cluster_6	U.S. Large Cap Growth	2026-08-06	11	-0.9%	0.16	0.75
MGK	cluster_6	U.S. Large Cap Growth	2026-08-05	12	-1.2%	0.21	0.75
IWF	cluster_6	U.S. Large Cap Growth	2026-08-06	11	-0.9%	0.16	0.75
QTEC	cluster_6	U.S. Large Cap Growth	2026-08-10	9	-0.5%	0.05	0.74
QQQ	cluster_6	U.S. Large Cap Growth	2026-08-07	10	-1.3%	0.09	0.74
VUG	cluster_6	U.S. Large Cap Growth	2026-08-05	12	-1.4%	0.23	0.73
PTNQ	cluster_6	U.S. Large Cap Growth	2026-08-07	10	-1.4%	0.09	0.73
XLK	cluster_6	U.S. Large Cap Growth	2026-08-07	10	-2.5%	0.13	0.73
IUSG	cluster_6	U.S. Large Cap Growth	2026-08-05	12	-1.6%	0.23	0.73
SPYG	cluster_6	U.S. Large Cap Growth	2026-08-05	12	-1.7%	0.24	0.72
VOOG	cluster_6	U.S. Large Cap Growth	2026-08-05	12	-1.7%	0.24	0.71
IVW	cluster_6	U.S. Large Cap Growth	2026-08-05	12	-1.7%	0.24	0.71
IWY	cluster_6	U.S. Large Cap Growth	2026-08-07	10	-1.8%	0.17	0.71
IYW	cluster_6	U.S. Large Cap Growth	2026-08-06	11	-1.3%	0.19	0.69
QTUM	cluster_6	U.S. Large Cap Growth	2026-08-13	6	-4.9%	0.06	0.67
ILCG	cluster_6	U.S. Large Cap Growth	2026-08-06	11	-1.5%	0.22	0.62
XNTK	cluster_6	U.S. Large Cap Growth	2026-08-13	6	-3.2%	0.05	0.59
IGM	cluster_6	U.S. Large Cap Growth	2026-08-06	11	-1.7%	0.24	0.56
NULG	cluster_6	U.S. Large Cap Growth	2026-08-05	12	-1.9%	0.25	0.54
IPO	cluster_6	U.S. Large Cap Growth	2026-08-13	6	-4.9%	0.11	0.29
PWB	cluster_6	U.S. Large Cap Growth	2026-08-07	10	-2.3%	0.25	0.12
PTF	cluster_6	U.S. Large Cap Growth	2026-04-07	95	10.0%	0.09	0.11
VPL	cluster_22	Japan Equity	2026-08-10	9	2.3%	0.01	0.97
IPAC	cluster_22	Japan Equity	2026-08-03	14	2.5%	0.14	0.86
BBJP	cluster_22	Japan Equity	2026-08-05	12	0.1%	0.15	0.83
EWJ	cluster_22	Japan Equity	2026-08-05	12	0.0%	0.16	0.82
DBJP	cluster_22	Japan Equity	2026-08-10	9	-1.1%	0.20	0.58
HEWJ	cluster_22	Japan Equity	2026-08-07	10	-0.8%	0.21	0.55
FLJH	cluster_22	Japan Equity	2026-08-10	9	-0.7%	0.20	0.53
EWJV	cluster_22	Japan Equity	2026-07-06	34	2.5%	0.66	0.34

DXJ	cluster_22	Japan Equity	2026-08-05	12	0.7%	0.47	0.27
TUR	cluster_20	India Equity	2026-08-14	5	2.9%	0.00	1.00
EWM	cluster_20	India Equity	2026-07-30	16	1.4%	0.05	0.95
REM	cluster_2	U.S. Mid Cap Equity	2026-08-12	7	0.2%	0.01	0.99
IWC	cluster_2	U.S. Mid Cap Equity	2026-08-05	12	0.8%	0.01	0.99
VBK	cluster_2	U.S. Mid Cap Equity	2026-08-07	10	-0.5%	0.01	0.98
SCHM	cluster_2	U.S. Mid Cap Equity	2026-08-07	10	0.2%	0.01	0.97
SMMV	cluster_2	U.S. Mid Cap Equity	2026-06-25	40	5.7%	0.04	0.96
VXF	cluster_2	U.S. Mid Cap Equity	2026-08-05	12	0.7%	0.04	0.96
ISCG	cluster_2	U.S. Mid Cap Equity	2026-08-04	13	-0.2%	0.04	0.96
SMLF	cluster_2	U.S. Mid Cap Equity	2026-08-03	14	1.7%	0.04	0.95
VTWO	cluster_2	U.S. Mid Cap Equity	2026-08-04	13	-0.6%	0.05	0.93
IWM	cluster_2	U.S. Mid Cap Equity	2026-08-04	13	-0.6%	0.05	0.93
IWO	cluster_2	U.S. Mid Cap Equity	2026-08-06	11	0.8%	0.03	0.91
VTWG	cluster_2	U.S. Mid Cap Equity	2026-08-06	11	0.7%	0.04	0.90
IVOG	cluster_2	U.S. Mid Cap Equity	2026-08-06	11	0.2%	0.08	0.88
MDYG	cluster_2	U.S. Mid Cap Equity	2026-08-06	11	0.3%	0.08	0.88
IJK	cluster_2	U.S. Mid Cap Equity	2026-08-06	11	0.3%	0.08	0.88
SCHA	cluster_2	U.S. Mid Cap Equity	2026-08-07	10	0.0%	0.01	0.85
IJS	cluster_2	U.S. Mid Cap Equity	2026-04-10	92	12.7%	0.37	0.62
ISMD	cluster_2	U.S. Mid Cap Equity	2026-04-09	93	17.3%	0.62	0.09
IYT	cluster_2	U.S. Mid Cap Equity	2026-08-06	11	0.9%	0.54	-0.45
KURE	cluster_13	China Equity	2026-07-06	34	11.5%	0.00	1.00
CXSE	cluster_13	China Equity	2026-08-07	10	-2.1%	0.31	0.60
GXC	cluster_13	China Equity	2026-07-31	15	0.0%	0.69	0.30
KBA	cluster_13	China Equity	2026-08-12	7	-1.3%	0.68	-0.55
KCE	cluster_14	Financials	2026-07-07	33	8.2%	0.01	0.99
FNCL	cluster_14	Financials	2026-04-13	91	11.3%	0.76	0.24
XLF	cluster_14	Financials	2026-04-13	91	11.3%	0.82	0.18
BSCQ	cluster_5	Aggregate Fixed Income	2026-08-07	10	0.1%	0.00	1.00
VGSH	cluster_5	Aggregate Fixed Income	2026-08-17	4	-0.0%	0.01	0.99
SHY	cluster_5	Aggregate Fixed Income	2026-08-17	4	0.0%	0.01	0.98
SCHO	cluster_5	Aggregate Fixed Income	2026-08-17	4	0.0%	0.01	0.97
LDUR	cluster_5	Aggregate Fixed Income	2026-08-17	4	-0.0%	0.03	0.96
SPTS	cluster_5	Aggregate Fixed Income	2026-08-17	4	-0.0%	0.03	0.93
SPSB	cluster_5	Aggregate Fixed Income	2026-08-18	3	0.0%	0.02	0.92
LMBS	cluster_5	Aggregate Fixed Income	2026-08-19	2	-0.4%	0.01	0.82
IBDR	cluster_5	Aggregate Fixed Income	2026-08-20	1	0.1%	0.32	0.38
PFXF	cluster_10	High Yield Bond	2026-08-10	9	-0.3%	0.12	0.86
SHYG	cluster_10	High Yield Bond	2026-08-19	2	-0.0%	0.00	0.74
HYLS	cluster_10	High Yield Bond	2026-08-13	6	-0.7%	0.10	0.59
HYG	cluster_10	High Yield Bond	2026-08-19	2	-0.1%	0.03	0.47
HYLB	cluster_10	High Yield Bond	2026-08-19	2	-0.1%	0.06	0.34
PFF	cluster_10	High Yield Bond	2026-08-13	6	-1.4%	0.53	0.13
SUB	cluster_16	Municipal Bond	2026-08-13	6	-0.0%	0.01	0.91

PSCC	cluster_18	Consumer Staples	2026-06-10	50	9.8%	0.26	0.74
PBJ	cluster_18	Consumer Staples	2026-08-18	3	1.1%	0.15	0.74
XLP	cluster_18	Consumer Staples	2026-07-27	19	0.7%	0.42	0.55
FSTA	cluster_18	Consumer Staples	2026-07-28	18	-1.5%	0.46	0.51
VDC	cluster_18	Consumer Staples	2026-07-28	18	-1.6%	0.46	0.49
XLRE	cluster_15	Real Estate	2026-08-18	3	1.0%	0.86	-0.54
VNQ	cluster_15	Real Estate	2026-08-19	2	-0.1%	0.90	-0.73
FREL	cluster_15	Real Estate	2026-08-19	2	-0.2%	0.91	-0.74
REET	cluster_15	Real Estate	2026-08-18	3	0.5%	0.99	-0.97

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
IBB	cluster_11	Healthcare	1.00	1.00
GSY	cluster_24	Cash / Ultra-Short Fixed ...	1.00	1.00
FXC	cluster_7	Asia-Pacific Equity	1.00	1.00
BSCQ	cluster_5	Aggregate Fixed Income	1.00	1.00
NANR	cluster_12	Materials	1.00	1.00
GDX	cluster_17	Gold	1.00	1.00
BKLN	cluster_23	Bank Loan / Floating Rate	1.00	1.00
SCHD	cluster_9	U.S. Dividend Equity	1.00	1.00
EQAL	cluster_3	U.S. Large Cap Value	1.00	1.00
ARKK	cluster_6	U.S. Large Cap Growth	1.00	1.00
FV	cluster_1	U.S. Large Cap Equity	1.00	1.00
COMT	cluster_19	Broad Commodities	1.00	1.00
SCHC	cluster_4	Developed ex-US Equity	1.00	1.00
EEMV	cluster_8	Emerging Markets Equity	1.00	1.00
XMHQ	cluster_2	U.S. Mid Cap Equity	1.00	0.99
KCE	cluster_14	Financials	1.00	0.99
VPL	cluster_22	Japan Equity	0.97	0.97
SUB	cluster_16	Municipal Bond	0.92	0.91
PFXF	cluster_10	High Yield Bond	0.99	0.86
PSCC	cluster_18	Consumer Staples	1.00	0.74

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
PSK	cluster_10	High Yield Bond	1.00	-1.00
UUP	cluster_26	Leveraged / Inverse	1.00	-1.00
IGF	cluster_7	Asia-Pacific Equity	1.00	-1.00
XTN	cluster_2	U.S. Mid Cap Equity	0.99	-0.99
RWO	cluster_15	Real Estate	0.99	-0.98
BNDX	cluster_5	Aggregate Fixed Income	0.98	-0.98

IGHG	cluster_23	Bank Loan / Floating Rate	0.93	-0.93
SPLV	cluster_9	U.S. Dividend Equity	0.98	-0.87
XLU	cluster_18	Consumer Staples	0.87	-0.87
EPHE	cluster_20	India Equity	1.00	-0.85
HYMB	cluster_16	Municipal Bond	0.79	-0.79
CQQQ	cluster_13	China Equity	0.71	-0.71
IAT	cluster_14	Financials	0.97	-0.70
CGW	cluster_3	U.S. Large Cap Value	0.92	-0.69
PSCT	cluster_6	U.S. Large Cap Growth	0.67	-0.63
PRN	cluster_1	U.S. Large Cap Equity	0.60	-0.59
IHF	cluster_11	Healthcare	0.87	-0.49
TDTT	cluster_21	Inflation-Protected Fixed...	0.35	-0.34
UNG	cluster_19	Broad Commodities	0.08	-0.08
AML	cluster_12	Materials	0.91	0.09

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.488	0.073	0.00016	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_20	10	FLAT	0.302	0.094	0.00021	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.730	0.096	0.00024	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.647	0.206	0.00036	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.759	0.028	0.00007	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_19	11	FLAT	0.591	0.168	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.734	0.119	0.00031	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.734	0.141	0.00039	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.715	0.181	0.00044	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.694	0.166	0.00045	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.699	0.187	0.00046	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_20	10	LONG	0.511	0.216	0.00055	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	58	LONG	0.847	0.125	0.00037	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.883	0.026	0.00001	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.883	0.054	0.00018	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.877	0.167	0.00032	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	LONG	0.819	0.201	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	58	LONG	0.833	0.239	0.00049	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_20	10	FLAT	0.447	0.270	0.00074	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.878	0.250	0.00047	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_20	10	FLAT	0.445	0.328	0.00083	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	LONG	0.781	0.275	0.00073	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.878	0.265	0.00055	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_20	10	FLAT	0.499	0.350	0.00101	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	17	LONG	0.742	0.352	0.00084	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	LONG	0.853	0.255	0.00073	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	LONG	0.867	0.322	0.00071	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_20	10	FLAT	0.559	0.408	0.00108	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	FLAT	0.903	0.237	0.00064	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.919	0.233	0.00059	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20